

OmniCRM Frontend

Engineering Brief

Complete Frontend Build via Figma Make
Static Data + Mock API Layer · Backend-Ready

Audience: Frontend Engineering Team
Author: Project Management
Baseline: Nightangle CRM v0 (170 pages, ~90,800 LOC)
Target: OmniCRM PRD v1.0 (16 modules, 100 user stories)

1. Executive Summary

1.1 Mission

Deliver a complete, production-quality OmniCRM frontend built in Figma Make, capable of running end-to-end customer journeys with realistic mock data — ready for backend integration. The frontend must demonstrate Salesforce/Zoho-class no-code customization (custom objects, fields, relationships, formulas, validation rules, layouts per role, custom apps), AI-led omni-channel engagement (Voice + WhatsApp + SMS + Email), node-based workflow automation, and India-primary compliance (GST, DLT, DPDP).

1.2 Why This Brief Exists

The current Nightangle CRM v0 codebase has strong UI surface coverage (~64% of PRD user stories have some UI), but suffers from four structural gaps that make it impossible to demo, test, or hand off to backend without rework:

- Module 3 (Customization & Metadata Platform) is missing entirely.** The platform's stated top differentiator does not exist as a UI. Every other module hardcodes its schema.
- Geography mismatch.** The codebase defaults to Nepal/NPR. PRD targets India-primary. This requires a global locale rework.
- Mock data is fragmented and incomplete.** Each hook seeds its own data; cross-module references break. There is no single source of truth for tenants, users, leads, deals, AI agents.
- 12 routes resolve to a Placeholder component.** These are explicit gaps that must be designed and built.

1.3 Scope of Work

Dimension	Baseline (v0)	Target (v1)
Pages	170	~210 (40 new)
Routes	181 (12 stubs)	230+ (no stubs)
User stories with UI	64 of 97	97 of 97
Acceptance criteria covered in UI	~15-20%	≥ 90%
Modules	12 (M3 missing, M13/M16 thin)	16 (PRD-aligned)
Geography	Nepal-primary	India-primary + GDPR-ready
Data architecture	Per-hook in-memory state	Centralized mock store with referential integrity

Customization depth	Lead form only	Salesforce-class Level 4
Workflow builder	Calls only (ConversationFlowEditor)	General-purpose multi-channel node builder
i18n readiness	Hardcoded English	Externalized strings, i18n framework wired

1.4 What 'Done' Looks Like

A frontend engineer should be able to deliver this milestone when, with mock data alone, all of the following are true:

- Every PRD user story has a corresponding UI flow that can be navigated end-to-end.
- Every acceptance criterion has a visible, demonstrable UI element (input, button, state, message, validation, transition).
- A new tenant can be created, a user added, a custom object designed, custom fields added with formulas and validations, a layout configured per role — and that custom object then appears in lead-related screens, in workflow builders, and in report builders.
- A campaign can be designed in the node builder spanning AI Call → WhatsApp → SMS → Email steps, and its execution can be observed in the campaign monitor.
- Every Super Admin destructive action (kill-switch, suspension, override, impersonation) requires confirmation, mandatory reason, and audit log entry — visible in the audit log viewer.
- All data flows through a single typed mock store. Backend can replace it with real APIs by swapping the data layer only — UI components do not change.
- All India-compliance surfaces (GSTIN, DLT, DPDP DSR, INR billing) are present and functional at the UI level.
- All routes resolve to real screens. Zero Placeholder components.

1.5 Out of Scope (Frontend Phase)

- Backend implementation, real APIs, real authentication, real persistence.
- Real telephony, WhatsApp Business API, SMS gateway, ESP, payment gateway integrations. (UI shows mock states.)
- Real AI inference. (Mock LLM responses with realistic latency.)
- Native mobile apps (iOS/Android) — responsive web only.
- Performance optimization beyond 'reasonable' (no need to load-test 10M leads — mock has 200).
- Accessibility deep audit (basic semantic HTML and keyboard navigation only — full WCAG audit is post-backend).

1.6 Key Numbers

Metric	Value	Notes
Total user stories to cover	97	From PRD §5 (Modules 1–16)
Total acceptance criteria	485	Each AC must be visible in UI
Estimated story points	~487	Based on gap analysis
Estimated calendar duration	10–14 weeks	1–2 frontend engineers + designer
Build waves	6	Per PRD §8 (Foundation → Platform Ops)
New pages required	~40	Including all M3 customization, M13 integrations, M16 DSR
Pages requiring rework	~85	Validation, India compliance, mock-store wiring
Pages reusable as-is	~45	Mostly UI shells with strong design

2. Methodology & Working Agreement

2.1 Engineer Workflow (Per User Story)

Every user story in this brief follows the same execution pattern. Treat each as a self-contained ticket.

5. **Read the user story and acceptance criteria** (linked to PRD §5).
6. **Locate the existing page or component** (file path provided in the brief). If none, scaffold a new one in the correct directory.
7. **Connect to the centralized mock store** (Section 4 of this brief). Never seed local data inside a component.
8. **Implement the UI to satisfy every AC** — validation, state transitions, error handling, loading states, success confirmations.
9. **Wire navigation and CRUD actions** to the mock store mutators (which simulate API latency and emit notifications).
10. **Add the route** if new, and confirm Sidebar / TopBar links lead to it.
11. **Self-verify with the AC checklist** (every AC = a clickable demonstration).
12. **Mark the story complete in tracker** (Linear sheet from the gap analysis).

2.2 Definition of Done (Per Story)

- UI implements every acceptance criterion (test by clicking through; each AC is demonstrable).
- All form fields have inline validation with helpful error messages.
- Loading state shown for async operations (use store's simulated latency).
- Success state shown with confirmation (toast or banner) and routes the user to the next logical page.
- Error states handled gracefully (retry button, fallback UI).
- Empty states designed (no records yet → friendly call-to-action).
- Permission gates enforced (page hidden / read-only based on role).
- Mobile responsive at 768px and 1024px breakpoints (desktop-first, mobile-friendly).
- Audit log entry written for any destructive or privileged action.
- Story documented in the changelog (mark which AC the change addresses).

2.3 Definition of Done (Per Module)

- All user stories complete (per above).
- Module dashboard / overview page exists and reflects mock data.
- Sidebar navigation includes all module pages, role-gated.
- Cross-module references work (e.g., a lead links to deals, deals link to campaigns, campaigns link to AI calls).

- Module-level analytics tile reflects in the global Reports & Analytics dashboard.
- Demo script: one walkthrough that exercises every primary flow in the module.

2.4 Working Agreement

Topic	Convention
Branch naming	feat/m{module}-{us-id}-{slug} (e.g., feat/m1-us-m1-2-license-assign)
PR scope	One user story per PR. Each PR must be reviewable in <30 minutes.
PR checklist	Linked to AC. Self-tested. Screenshots of each AC. Mock data updated.
Code review	Required before merge. Reviewer checks AC → UI mapping.
Design tokens	Use existing CSS variables in default_shadcn_theme.css. Do not introduce new colors.
Component library	Reuse from src/app/components/ui/ first. Add new only if no fit.
State management	Centralized mock store (Section 4). No useState for cross-page data.
Routing	Update routes.tsx in the same PR as the page.
Testing	Manual QA via mock data. Automated tests deferred to backend phase.
Accessibility	Semantic HTML, keyboard navigation, focus rings. Full WCAG audit post-backend.

2.5 Tools & Stack

Layer	Tooling
Build	Figma Make (Vite + React 18 + TypeScript)
Routing	react-router 7
Styling	Tailwind 4 + shadcn theme (default_shadcn_theme.css)
UI primitives	Radix UI (already installed, 23 packages)
Components	MUI 7 (heavy use) + custom shadcn-style in components/ui/
Icons	lucide-react + @mui/icons-material
Charts	Recharts (existing) — sufficient for v1
Forms	React Hook Form + Zod (recommend introducing — currently bare useState)
Tables	@tanstack/react-table (recommend introducing for sortable lists)
Dates	date-fns (already installed)

Notifications	sonner (already installed) — toast pattern
Drag-drop	@dnd-kit/* (already installed) — for kanban + workflow builder
Mock data	Centralized in src/app/store/ (new — see Section 4)
i18n	react-i18next (introduce in Wave 1)

3. Success Criteria & Acceptance

3.1 Module-Level Acceptance

A module is accepted when all of these are true on the running mock-data build:

Module	Acceptance Walkthrough
M1 Tenant & License	Create tenant w/ KYC → see in tenant list → assign license w/ maker-checker → suspend → restore. All actions logged in audit log.
M2 User, Role & Permission	Create user → assign roles → custom-role builder shows custom objects from M3 → 2FA enrollment flow → force logout.
M3 Customization & Metadata	Create custom object 'Distributor' → add Lookup field to Lead → add Formula field → add validation rule → design layout per role → publish → see new fields appear in Lead detail page.
M4 Lead Management	Import 100 leads via CSV → assign via round-robin → score → detect duplicates → merge → view timeline w/ AI badge.
M5 Deal & Pipeline	Create deal from lead → add line items → trigger discount approval → view forecast → close as Won w/ reason → see attribution.
M6 AI Calling	Configure AI agent → design conversation flow (node-based) → schedule outbound call → live monitor → warm transfer to human → review recording w/ consent log.
M7 WhatsApp	Submit template for Meta approval → run broadcast w/ caps → AI replies → supervisor takeover → opt-out enforced.
M8 SMS	Register DLT template → schedule promotional campaign respecting quiet hours → opt-out → reminder + reply handling.
M9 Email	Inbound email creates lead → AI drafts reply → drip sequence → unsubscribe → deliverability dashboard.
M10 Campaigns & Workflow	Build node-based workflow Call → WhatsApp → SMS → Email → execute → A/B test → pause → resume → see budget cap pause.
M11 Business Playground	Run AI research → ingest KB doc → generate content → schedule social post → scrape leads → see credit debit.
M12 Analytics	Build custom dashboard → drag-drop tiles → drill into AI metrics → schedule report → see anomaly insight card.
M13 Integrations	Assign phone number → connect Google Calendar → configure outbound webhook w/ HMAC → connect Meta Lead Ads → export to accounting.
M14 Billing	Top-up via Stripe → see GSTIN invoice → offline payment proof + dual approval → trigger soft alert → manual credit adjustment.
M15 Admin Controls	Trigger kill-switch w/ 2FA + reason → grant time-bounded override → impersonate tenant user → onboard telephony provider w/ test calls.

M16 Security & Compliance

Search audit log → fulfill DSR (export + erase) → configure AI guardrails → trigger incident response → backup + restore preview.

3.2 Cross-Cutting Acceptance

- **Single mock store:** every page reads/writes through the centralized store. Confirmed by grepping for useState with shared data — should return zero.
- **Referential integrity:** deleting a tenant cascades correctly; deleting a user reassigns their leads; deleting a custom object warns about dependent fields/layouts/workflows.
- **India compliance visible:** INR throughout, GSTIN on invoices, DLT on SMS, DSR on audit, PAN format validation.
- **Audit log completeness:** every privileged action writes a log entry. Visible in audit viewer with actor, timestamp, before/after, reason.
- **Permission enforcement:** auditor role sees no edit buttons; sales agent cannot see billing; tenant admin cannot impersonate; super admin sees all but actions are gated.
- **AI attribution:** every AI action displays the AI badge with agent ID + campaign ID, never linked to a human user.
- **Maker-checker pattern:** license assign, credit adjustment, offline payment, override grant, impersonation — all show pending → approve → execute flow.

3.3 Final Demo Script

The build is complete when this 30-minute demo runs cleanly:

13. Super Admin logs in with 2FA. Reviews platform health dashboard.
14. Onboards a new tenant (TechAvtar India Pvt Ltd) with PAN, GSTIN, KYC documents. Watches provisioning state machine. Confirms welcome notifications.
15. Assigns a license. Maker-checker approval flow. License becomes Active.
16. Switches context to tenant (impersonation, time-bounded). Tenant Admin view loads.
17. In tenant context: creates a custom object 'Distributor' with 8 fields including a formula and a lookup to Account. Designs Sales Agent and Manager layouts. Publishes.
18. Imports 100 mock leads via CSV with field mapping. Sees deduplication. Assigns via round-robin to 5 agents.
19. Designs a campaign in the node-based workflow builder: AI Call → if no answer → WhatsApp → if reply 'YES' → create Deal → assign Sales Agent.
20. Submits a WhatsApp template for Meta approval. Approves it. Activates the campaign.
21. Watches campaign execution monitor: AI calls in flight, WhatsApp messages going out, leads progressing through stages.
22. A deal closes Won. Triggers attribution view showing contributing campaigns and AI calls.
23. Opens billing: sees credit consumption, GSTIN-compliant invoice generated, soft alert at 80% threshold.

24. Returns to Super Admin. Reviews audit log: every action attributable. Triggers a (mock) data subject request — generates export.
25. Triggers a (mock) emergency kill-switch with 2FA + reason. All AI calls in fleet halt. Confirmation logged.

4. Mock Data & State Architecture

4.1 Why a Centralized Mock Store

The current codebase has each hook seed its own data — useLeads has its leads, useDeals has its deals, useCampaigns has its campaigns. They never reference each other. A lead in useLeads and a deal in useDeals about the same prospect have no link. This breaks every cross-module flow.

Required pattern: a single in-memory mock store with referential integrity, simulated latency, and a contract identical to what the real backend will eventually expose. The hooks become thin wrappers that call the store. Backend swap = replace store implementation. Components do not change.

4.2 Store Structure (Recommended)

Place under `src/app/store/`. Use Zustand for the store engine (lightweight, no provider needed) — install if not present.

```
src/app/store/ |— index.ts           // root store with all slices |— types.ts
// shared TypeScript types (entities, IDs) |— seed.ts           // initial seed data
(200 leads, 50 deals, 20 users, etc.) |— api.ts               // simulated API: latency,
errors, pagination |— slices/ |— tenants.ts                 // tenants + licenses |—
users.ts           // users + roles + permissions + sessions |— metadata.ts           //
custom objects + fields + layouts + apps ← M3 |— leads.ts      // leads +
timelines + duplicates |— deals.ts                          // deals + line items + stages +
forecasting |— aiCalls.ts // calls + agents + flows + transcripts |—
whatsapp.ts       // messages + templates + campaigns + consent |— sms.ts
// messages + DLT templates + opt-outs |— email.ts           // messages + templates +
suppression |— campaigns.ts // campaigns + workflows + execution logs |—
playground.ts     // research sessions + KB docs + assets + content |— analytics.ts
// dashboards + reports + saved views |— integrations.ts      // numbers + webhooks +
calendars + providers |— billing.ts // invoices + ledger + credits + payments |—
|— adminControls.ts // kill-switches + overrides + impersonation |—
auditCompliance.ts // audit log + DSR + incidents + backups |— hooks/           //
public API for components |— useTenants.ts |— useUsers.ts |— useMetadata.ts
|— useLeads.ts |— ... (one per slice)
```

4.3 Cross-Slice References

Every entity has a global ID. Cross-references use IDs, not embedded objects. Example:

```
// types.ts export type TenantId = `TNT-${string}`; export type UserId = `USR-${string}`; export
type LeadId = `LD-${string}`; export type DealId = `DL-${string}`; export type CampaignId =
`CMP-${string}`; export type AICallId = `CALL-${string}`; export interface Lead { id: LeadId;
tenantId: TenantId; ownerId: UserId | null; primaryPhone: string; // ... customFields:
Record<string, unknown>; // values keyed by Module 3 field API names relatedDealIds:
DealId[]; relatedCampaignIds: CampaignId[]; relatedCallIds: AICallId[]; timeline:
TimelineEntry[]; // append-only}
```

4.4 Simulated API Layer

The `api.ts` module wraps every store mutator to simulate the real backend's behavior:

- **Latency:** 300–800 ms for reads, 600–1500 ms for writes. Configurable per call. UI must show loading states.
- **Failure injection:** 5% random failure rate (configurable). UI must show retry-friendly errors.
- **Pagination:** list endpoints return { items, total, page, pageSize, hasMore }. Default pageSize = 25.
- **Optimistic updates:** supported via the store's transaction helper.
- **Audit log writes:** every privileged mutator emits an audit entry automatically.
- **Notifications:** the store dispatches Sonner toasts on success/failure.
- **WebSocket simulation:** for live feeds (call monitor, campaign execution, dashboards), use a setInterval-based pub/sub. UI subscribes via useEffect.

4.5 Seed Data Requirements

The seed must be rich enough to demonstrate every flow without requiring the demo operator to first create dozens of records.

Entity	Seed Count	Notes
Tenants	12	Mix of statuses: 6 Active, 2 Provisioning, 2 Suspended, 1 Locked, 1 Trial. Indian companies with realistic GSTIN, PAN, addresses.
Users (per tenant)	5–15	Mix of roles: 1 Tenant Admin, 2 Managers, 5–10 Sales Agents, 1–2 Support Agents, 1 Auditor.
Leads	200 per active tenant	Mix of stages, sources, scores. Some duplicates for dedup demo. Some with rich timelines (10+ events).
Custom objects	3 per tenant	Distributor, Service Contract, Warranty Claim — pre-built so M3 demo doesn't start from zero.
Deals	50 per active tenant	Across pipelines + stages. Some stalled, some won, some with discount approvals pending.
AI Agents	5 per tenant	Sales, Support, Collections, Survey, Renewal.
Conversation flows	3 per agent	Pre-built so designer can demo flow execution.
WhatsApp templates	10 per tenant	Mix: 6 approved, 2 pending, 2 rejected. Various categories.
SMS DLT templates	8 per tenant	Mix: transactional, promotional, service-implicit. Mix of approved/pending.
Email templates	10 per tenant	
Campaigns	15 per tenant	Mix of running, paused, completed, scheduled.
AI Call records	300 per tenant	Across last 30 days. Mix of outcomes.

Audit log entries	1000 per tenant	Spanning user actions, AI actions, system actions.
Invoices	12 per tenant	Last 12 months. Mix of paid, pending, overdue.
Phone number inventory	100 platform-wide	Mix of regions (India, Nepal, US). Some assigned, some available.

4.6 Store API Contract (Standardized)

Every slice exposes the same shape so the conversion to real backend is mechanical:

```
// Standard slice contract
export interface SliceAPI<T, CreateT, UpdateT> {
  // Reads
  list(params?: ListParams): Promise<Page<T>>;
  get(id: string): Promise<T | null>;
  search(query: string): Promise<T[]>;
  // Writes (each writes audit log)
  create(input: CreateT, opts?: WriteOpts): Promise<T>;
  update(id: string, input: UpdateT, opts?: WriteOpts): Promise<T>;
  remove(id: string, reason: string): Promise<void>;
  // Subscriptions (for live updates)
  subscribe(callback: (event: ChangeEvent<T>) => void): UnsubscribeFn;
}

// Standard write options
export interface WriteOpts {
  reason?: string; // required for destructive actions
  approverIds?: UserId[]; // required for maker-checker actions
  triggerNotification?: boolean;
  audit?: AuditMeta;
}
```

Note: The store is the single source of truth. Components must NEVER seed data inline. PR review will reject any `useState` containing entity data that should live in the store.

4.7 Backend Swap Path

When backend is ready, the swap should require touching only `api.ts` and nothing else:

26. Replace `api.ts` with real fetch / axios calls.
27. Keep slice contract identical.
28. Replace WebSocket simulation with real SSE / WebSocket.
29. Components, pages, hooks remain unchanged.
30. Mock store moves to a 'demo mode' toggle for showcasing.

5. Information Architecture

5.1 Three-Tier Application Shell

The platform serves three distinct user tiers, each with its own shell, sidebar, and theme accent. The current codebase already has admin / customer / superadmin folders — keep this separation, formalize the layouts.

Tier	Audience	Shell Path	Sidebar Theme
Super Admin	Hitech Solutions staff	/admin/*	Deep navy (#0D2840)
Tenant (Customer)	End customers (TechAvtar etc.)	/tenant/* or /:tenantSlug/*	Brand blue (#1F4E79)
Customer Portal (Tier 3)	Distributors of tenants	/portal/*	Light theme — limited surface

Note: the current code uses /admin/* for Super Admin and root /* for tenant. Keep that pattern but formalize: Super Admin pages MUST go under /admin, tenant pages MUST go under tenant-scoped routes. Customer Portal (Tier 3) is new — see Section 6 module map.

5.2 Sidebar Navigation Structure

Super Admin Sidebar

Section	Items
Overview	Dashboard · Platform Health
Tenants	All Tenants · Create Tenant · License Management · Subscription Plans
Operations	Telephony Inventory · Provider Onboarding · Number Pool
Billing & Finance	Platform Billing · Tenant Invoices · Credit Management · Payment Reconciliation
Controls	Emergency Controls · Overrides · Impersonation Sessions · Maintenance Mode
Compliance	Audit Explorer · DSR Queue · Incident Response · Backup & DR
Analytics	Cross-Tenant Analytics · Revenue · System Health
Settings	Platform Policies · Security · API Keys · Integrations

Tenant (Customer) Sidebar

Section	Items
Overview	Home · My Dashboard
CRM	Leads · Deals · Pipeline · Companies · Contacts · Activities

Custom Apps	(dynamically populated from M3 — e.g. 'Distributor Management')
AI Calling	Dashboard · Live Monitor · Call Logs · AI Agents · Conversation Flows · Transfer Queue · Cost Governance · Compliance
WhatsApp	Inbox · Campaigns · Templates · Automation · Consent · Analytics · Settings
SMS	Inbox · Campaigns · Templates · Short Links · DLT Registry · Analytics · Settings
Email	Inbox · Compose · Campaigns · Templates · Signatures · Deliverability · Settings
Campaigns & Workflows	All Campaigns · Workflow Builder · Execution Monitor · Budget Manager · A/B Tests
Business Playground	AI Research · Knowledge Base · Asset Library · Content Studio · Social Media · Lead Discovery
Reports & Analytics	Dashboards · Custom Dashboard Builder · Reports · Custom Report Builder · Anomaly Insights
Integrations	Calendar · Webhooks · Lead Sources · Accounting Export · API Keys
Billing	Overview · Usage · Invoices · Credit Purchase · Payment Methods
Settings	Company · Customization (M3) · Users & Roles · Security · Compliance · API

Customer Portal Sidebar (Tier 3 — Distributors)

Section	Items
Overview	Dashboard · My Account
Operations	Place Orders · Track Orders · Order History · Documents
Communications	Tickets · Messages
Settings	Profile · Notifications

5.3 Top Bar Pattern

Every page has a TopBar (already exists at `src/app/components/layout/TopBar.tsx`). Standard contents:

- Left: hamburger toggle for sidebar collapse + breadcrumb trail
- Center: global search (cmd+K) — searches across leads, deals, contacts, custom records
- Right: notification bell · help · user menu (profile, switch tenant for super admin, logout)
- Below: contextual actions for the current page (e.g., 'Create Lead', 'Bulk Actions')

5.4 Common Page Patterns (Template Reuse)

Every page in the platform fits into one of seven patterns. Reuse the same structure for consistency.

Pattern	Use For	Key Sections
List View	Leads, Deals, Users, Tenants, Invoices, Calls, Campaigns...	Filters · Saved Views · Bulk Actions · Table/Cards · Pagination
Detail View	LeadDetail, DealDetail, TenantDetail...	Header (name + status + actions) · Tabs (Overview, Activity, Related, Settings)
Form / Wizard	CreateTenant, CreateDeal, CreateCampaign...	Stepper · Per-step validation · Review summary · Submit
Builder Canvas	Workflow Builder, Conversation Flow Editor, Custom Object Builder, Layout Editor, Dashboard Builder	Sidebar (palette) · Canvas (drag-drop) · Inspector (selected element properties)
Inbox / Conversation	WhatsAppInbox, EmailInbox, SMSInbox	Conversation list · Active thread · Composer · Side panel (lead context)
Dashboard	AdminDashboard, CustomerDashboard, ReportsAnalytics	KPI tiles · Charts · Drill-down links · Date range picker · Refresh
Settings Page	Company, Security, Customization, Compliance...	Section navigation · Form sections · Save bar (sticky)

5.5 Routing Conventions

All routes use react-router 7. Conventions:

- List page: /entity (e.g., /leads, /deals)
- Create page: /entity/create
- Detail page: /entity/:id
- Edit page: /entity/:id/edit
- Sub-resource: /entity/:id/:sub (e.g., /campaigns/:id/monitor)
- Settings: /settings/:section
- Builder: /builder/:type/:id (e.g., /builder/workflow/wf_123)
- Super Admin: /admin/* prefix for ALL super admin routes

Inconsistencies in current codebase to fix: deals/list and deals (both exist), some routes use kebab-case, some use snake. Standardize to kebab-case throughout.

5.6 Theme & Design Tokens

Use the existing default_shadcn_theme.css as the source of truth. Do not introduce new colors. Reference variables, never hardcoded hex.

Token	Use For
--primary	Primary action buttons, brand emphasis
--secondary	Secondary surfaces
--destructive	Delete, Suspend, Kill-switch buttons
--success	Success toasts, Won deals, Active status
--warning	Warnings, Pending approvals, Soft alerts
--muted	Disabled states, helper text
--border	Card borders, dividers

Module 1 — Tenant & License Management

Mission: Super Admin onboards customers, manages licenses, enforces commercial terms. India-primary KYC. Maker-checker on critical actions.

Pages to Build / Refactor

Page	Path	Status	Notes
Tenant List	/admin/tenants	Reuse + filters	Add status filters (Active/Provisioning/Suspended/Locked/Trial), bulk actions, sortable columns.
Create Tenant Wizard	/admin/tenants/create	Already fixed (US-M1.1)	4-step wizard with India-primary defaults, KYC, document versioning, provisioning state machine.
Tenant Detail	/admin/tenants/:id	Refactor	Tabs: Overview · Users · License · Usage · Audit · Documents (with version history). Add suspend/restore/lock controls with mandatory reason.
Tenant Edit	/admin/tenants/:id/edit	Refactor	Same as create but edit mode. KYC field changes are versioned.
License Management	/admin/licenses	Reuse	List of all licenses across tenants. Filter by status/tenant/type.
License Detail	/admin/licenses/:id	Refactor	Add: maker-checker status, immediate-vs-next-cycle toggle, upgrade/degrade flow, temp license + payment proof flow.
Subscription Plans	/admin/subscription-plans	Reuse + India pricing	Replace NPR with INR throughout. Add GST tax-inclusive toggle.
Plan Detail / Edit	/admin/subscription-plans/:id	Reuse	
Usage Analytics	/admin/tenants/:id/usage	Refactor	Add real-time hard-stop indicator, threshold visualizer, propagation status.
Suspension Confirmation Modal	(modal)	NEW	Mandatory reason code (compliance/non-payment/fraud/other), confirmation, audit log preview.
License Approval Queue	/admin/licenses/approvals	NEW	Maker-checker queue: pending license assignments awaiting second approver.

Temp License + Payment Proof	/admin/licenses/:id/payment	NEW	Upload payment proof, dual-approver flow, auto-conversion to permanent on approval.
------------------------------	-----------------------------	-----	---

Customizations & Special Behavior

- **India-primary defaults:** INR currency, IST timezone, en-IN locale, +91 dial code, Indian states dropdown, PAN/GSTIN fields with format validation.
- **Document versioning:** every KYC document upload creates a new version. Previous versions are retained, never overwritten. View History drawer shows all versions chronologically.
- **Provisioning state machine:** Submitting → Provisioning DB → Configuring → Notifying → Active. Each state visible. Email + WhatsApp delivery confirmations during Notifying.
- **Maker-checker pattern:** license assignment requires two distinct approvers. UI shows: Maker submits → Pending Approval → Checker reviews → Approve/Reject. Both approvers logged.
- **Hard stop visualization:** tenant detail page shows live credit balance with threshold gauges (80%/90%/100%). At 100% the tenant card visibly switches to 'Hard Stop Active' state.
- **Upgrade vs degrade rules:** upgrade applies immediately or next cycle (toggle). Degrade always applies at end of current cycle (forced).
- **Audit visible:** every action on a tenant — create, license assign, suspend, restore, override — appears in the tenant's audit log tab with actor, timestamp, before/after, reason.

CRUD Operations

Action	Trigger	Mock Behavior
Create Tenant	Wizard final step	POST creates tenant in 'Provisioning' status. State machine animation runs. Audit log written.
Suspend Tenant	Suspend button on detail page	Modal asks for reason code + confirmation. Status flips to 'Suspended'. All active sessions terminated (mock toasts shown).
Restore Tenant	Restore button on suspended tenant	Status returns to 'Active'. Paused campaigns remain paused (manual resume required).
Assign License	From tenant detail or license create	Creates license in 'Pending Approval' status. Routes to approval queue. Second approver clicks 'Approve' → 'Assigned'.
Upgrade License	Upgrade button on license detail	Modal: pick new plan, choose Apply Immediately vs Next Cycle. Old license stays linked for audit.

Issue Temp License	From sales context	Creates license with 15-day expiry. Background timer counter visible on tenant card.
Approve Payment Proof	From temp license queue	Two-approver flow. On approval, license auto-converts to Permanent.

Navigation Connections

- Top: 'Create Tenant' button always visible on tenant list
- Tenant list rows are clickable → tenant detail
- Tenant detail Users tab → links to /admin/users (filtered by tenant)
- Tenant detail License tab → links to /admin/licenses/:id
- License detail → 'View Tenant' button → /admin/tenants/:id
- Approval queue badge in Super Admin sidebar shows pending count

Demo Walkthrough

Super Admin opens Tenants → Create Tenant. Fills KYC for an Indian company. Watches provisioning. Sees tenant in list (Active). Opens detail → Assigns license (Maker). Switches to second admin → Approves (Checker). License is Active. Suspends tenant with reason 'compliance'. Restores 5 minutes later. All actions visible in audit tab.

Existing Files (Reuse / Refactor)

src/app/pages/admin/CreateTenant.tsx (already fixed) · TenantList.tsx · TenantDetail.tsx · LicenseManagement.tsx · LicenseDetail.tsx · SuperAdminControls.tsx

Module 2 — User, Role & Permission Management

Mission: Tenant Admin manages users and roles. Permissions are metadata-driven (read from M3). Every AI action distinguished from human actions.

Pages to Build / Refactor

Page	Path	Status	Notes
Users List	/users (tenant) and /team (alt)	Reuse + filters	Active/Inactive/Pending filter, role filter, sortable columns.
Create / Edit User	/users/create, /users/:id/edit	Reuse + 2FA enrollment	Add 2FA enrollment trigger on first login, named-user-limit enforcement banner when at limit.
User Detail	/users/:id	Refactor	Tabs: Overview · Roles · Sessions · Activity · Audit. Add force-logout button, session list.
Roles List	/roles-permissions/roles	Reuse	List of all roles in the tenant. Clone, edit, delete actions.
Create / Edit Role	/roles-permissions/create, /:id/edit	Major refactor	Permission tree must be metadata-driven: read object list from M3 store, render checkboxes per object × per action × per role. Drag-drop seed roles.
Permission Matrix	/roles-permissions/matrix	Major refactor	Cross-tab view: rows = permissions, columns = roles. Cells are toggles. Live preview of which menu items each role sees.
User-Role Assignment	/roles-permissions/assignments	Reuse	
Permission Audit Log	/roles-permissions/audit	Reuse	
2FA Enrollment Wizard	/2fa/enroll	NEW	QR code scan + 6-digit verification + backup codes. Required when tenant has 2FA enforced.
Active Sessions Page	/users/:id/sessions	NEW	List of active sessions with device/IP/location, force-logout per-session, force-logout all.
Concurrent Session Limit Warning	(banner)	NEW	When at limit, login attempts show: 'Concurrent user limit reached (5/5). Terminate oldest session?' modal.

Customizations & Special Behavior

- **Metadata-driven permissions:** the permission tree is generated by reading custom objects from M3 store. When tenant adds a custom object 'Distributor', it automatically appears as 'Distributor' branch in the permission tree with View/Create/Edit/Delete checkboxes.
- **DENY-over-ALLOW resolution:** if a user has multiple roles and one denies a permission, UI shows 'Denied' with conflict info on hover.
- **Reserved permissions blocked:** Super Admin and Billing-Write checkboxes cannot be enabled by Tenant Admin. UI greys them out with explanation.
- **AI vs Human distinction:** every audit log entry shows actor type icon (person silhouette for human, robot for AI, gear for system). AI entries also show Agent ID + Campaign ID.
- **Force logout flow:** Tenant Admin opens user → Sessions tab → 'Force Logout' button → modal with reason → all sessions invalidated (mock). Audit log shows force-logout event with reason.
- **2FA optional toggle:** tenant settings has 'Enforce 2FA for all users' toggle. When ON, existing users see 2FA enrollment on next login. New users prompted on first login.

CRUD Operations

Action	Trigger	Mock Behavior
Create User	Users → New	POST → user in 'Pending' (set-password link emailed mock). Email toast appears. Audit logged.
Assign Roles	Edit user → Roles tab → multi-select	PATCH → roles updated. Permission preview re-renders showing new accessible menu items.
Force Logout	User detail → Sessions → Force Logout	Mock: all sessions invalidated. User card shows 'Logged Out 5 sec ago' with force-logout reason.
Deactivate User	User detail → Deactivate	Mandatory reason. Status flips. User cannot log in until reactivated.
Build Custom Role	Roles → New	Drag from seed role → modify → save. Permission tree reflects M3 custom objects.
Enroll 2FA	First login when 2FA enforced	QR code → enter 6-digit → success. Backup codes shown once.

Navigation Connections

- User list rows → user detail
- User detail → 'Send Reset Link' action
- Permission matrix → click any cell → opens role detail with that permission highlighted
- Audit log entries with AI actor → click → AI Agent detail (M6)

Demo Walkthrough

Tenant Admin creates user 'Priya' with Sales Agent role. Priya logs in, prompted for 2FA enrollment. Tenant Admin views Priya's session. Force-logs her out citing 'security review'. Audit log shows the action.

Existing Files (Reuse / Refactor)

*src/app/pages/admin/CreateEditRole.tsx · PermissionMatrix.tsx · UserRoleAssignment.tsx ·
src/app/pages/customer/TenantCreateEditRole.tsx · TenantPermissionMatrix.tsx*

Module 3 — Customization & Metadata Platform (Level 4 — Salesforce/Zoho-class)

Mission: The platform's top differentiator. Lets tenants design custom objects, fields, relationships, formulas, validation rules, layouts per role, and custom apps — without writing code. Every other module reads from this metadata.

Pages to Build / Refactor

Page	Path	Status	Notes
Customization Hub	/settings/customization	NEW	Landing page: cards for Objects, Layouts, Apps, Validation Rules. Stats: '12 custom objects · 87 custom fields · 4 apps'.
Object List	/settings/customization/objects	NEW	Grid/list of all objects (seed: Lead, Deal, Contact, Account + custom). Each card shows label, API name, field count, layout count. Actions: Edit, Clone, Delete.
Object Builder (Create)	/settings/customization/objects/create	NEW	Wizard: Label + Plural Label + API Name + Description + Allow Reports + Allow Activities. Validation: API name unique, no reserved names. Auto-generates default fields (ID, CreatedBy, CreatedAt, UpdatedBy, UpdatedAt).
Object Detail / Edit	/settings/customization/objects/:id	NEW	Tabs: Fields · Relationships · Layouts · Validation Rules · List Views · Triggers (workflow hooks). Schema viewer with relationship diagram.
Field Builder	/settings/customization/objects/:id/fields/create	NEW	Type picker first (20+ field types per PRD). Then type-specific config. Common: required, unique, default, help text, validation regex.
Formula Field Builder	/settings/customization/objects/:id/fields/formula	NEW	Spreadsheet-like expression editor with function picker (IF, AND, OR, TEXT, VALUE, DATE, CASE, ISBLANK, LENGTH, LEFT/RIGHT/MID), field reference picker (with dot-notation for cross-object via lookups), real-time syntax validation, circular reference detection.
Validation Rule Builder	/settings/customization/objects/:id/validations/create	NEW	Condition builder (field operator value AND/OR ...) + error message

			+ per-field placement. Rule preview against sample records.
Page Layout Editor	/settings/customization/objects/:id/layouts/:role	NEW	Drag-drop canvas: section panel (palette of fields + section dividers + related lists) → layout area. Per-field: visibility, read-only, required-on-this-layout. Save creates a new version.
List View Builder	(modal on object lists)	Refactor SavedView sSidebar	Generalize current Lead-only saved views to any object. Filter conditions + columns + sort + share scope (Private/Public/Role-specific).
Custom Apps List	/settings/customization/apps	NEW	Grid of custom apps. Each: name, included objects, assigned roles. Actions: Edit, Clone, Delete, Activate/Deactivate.
Custom App Builder	/settings/customization/apps/create	NEW	Wizard: name + icon + description → pick objects → arrange tabs → set default tab → assign roles. Preview shows app launcher card and sidebar.
Schema Viewer	/settings/customization/schema	NEW	Visual diagram of all objects and relationships (use react-flow or similar). Click any object to navigate to its detail.
Metadata Version History	/settings/customization/history	NEW	Last 30 schema versions. Diff viewer for any two versions. Rollback action with confirmation.
Field Type Picker	(modal/component)	NEW	Reusable component listing all 20+ field types with icons and descriptions: Text, Textarea, Number, Decimal, Currency, Percentage, Date, DateTime, Time, Email, Phone, URL, Checkbox, Picklist, Multi-Select Picklist, Attachment, Image, Lookup, Master-Detail, Formula, Rollup Summary, Auto-Number, Geolocation.

Customizations & Special Behavior

- **Metadata is THE source of truth:** every other module (Leads, Deals, Workflows, Reports, Permissions) reads its schema from the M3 store. Adding a custom field = it appears everywhere it should appear, no code changes.
- **Field type richness:** support all 20+ types from PRD. Each type has type-specific configuration UI (e.g., Picklist needs options list, Lookup needs target object picker, Formula opens the expression editor).

- **Formula engine UX:** real-time syntax check (red squiggle on errors), function autocomplete, field-reference autocomplete with dot-notation (e.g., Lead.Company.Industry), preview pane showing computed value for the first 5 sample records.
- **Validation rule UX:** condition builder with AND/OR groups, error message editor, 'Test against sample records' button shows which records would fail.
- **Layout editor:** two-column canvas (sections + fields) drag-drop. Per-role tabs at top. Live preview with mock record data. Field-level read-only / required / hidden toggles.
- **Soft-delete with 30-day recovery:** deleted fields/objects move to a Recycle Bin section. Restore action available for 30 days.
- **API name immutability:** after first save, API name field is locked (only label is editable). Migration warning if user attempts to delete a field with data.
- **Sandbox preview:** before publishing schema changes, 'Preview in Sandbox' shows how affected pages will look without affecting production.
- **Custom App = full workspace:** when a Custom App is published, it appears in the App Launcher (top-right TopBar dropdown). Switching to it changes the entire sidebar.

CRUD Operations

Action	Trigger	Mock Behavior
Create Object	Object Builder Save	Object created. Auto-generates default system fields. New object appears in Object List, Schema Viewer, Permission Tree, Workflow Trigger picker, Report Builder.
Add Field	Field Builder Save	Field added. Layouts updated to include it. Records get default value. Lead/Deal/etc. detail pages show the new field section.
Add Lookup Field	Field Builder → Type=Lookup → pick target	Relationship registered. Both objects' Schema Viewer reflects the connection. Lookup picker available in record forms.
Add Formula Field	Formula Builder Save	Server-side evaluation simulated. Computed values appear on record reads (lazy, not backfilled).
Save Layout	Layout Editor Save	Layout version created. Affected role's users see new layout on next page load.
Publish App	App Builder Save	App appears in App Launcher for assigned roles.
Soft-Delete Field	Field row Delete	Field marked deleted. Data preserved. Recovery available 30 days. Affected layouts show 'Field deleted' indicator.
Restore Field	Recycle Bin Restore	Field reactivated. Layouts updated.

Rollback Schema	Version History Rollback	Confirmation modal warns of impact. On confirm, schema reverts. Audit log entry.
-----------------	--------------------------	--

Navigation Connections

- Customization Hub is the entry point — link prominently from Settings sidebar
- Object Detail → Fields tab → New Field → Field Builder
- Object Detail → Layouts tab → role tabs → Layout Editor
- Schema Viewer is interactive — click an object to drill in
- Custom App Launcher in TopBar (top-right) shows all apps user has access to
- When user switches to a Custom App, sidebar replaces with that app's tabs

Demo Walkthrough

Tenant Admin opens Customization Hub. Creates 'Distributor' object with 8 fields including a Lookup to Account, a Formula 'DaysSinceLastOrder', and a validation rule 'Credit Limit must be > 0 when Status = Active'. Designs a Sales Agent layout (5 fields shown) and a Manager layout (all fields). Publishes a 'Distributor Management' app combining Distributor + Order + Shipment. Assigns app to Sales Agents. Sales Agent logs in → sees Distributor Management in App Launcher → opens it → sees only Sales Agent layout fields → tries to save with empty Credit Limit → validation error appears.

Existing Files (Reuse / Refactor)

BUILD FROM SCRATCH. The only existing reusable code is LeadFormConfiguration.tsx (413 L) — generalize its pattern to all objects. SavedViewsSidebar / CreateViewModal can be generalized for List View Builder.

Module 4 — Lead Management (CRM Core)

Mission: System of record for prospects. Reads schema from M3. Multi-source capture, scoring, dedup, unified timeline, bulk operations.

Pages to Build / Refactor

Page	Path	Status	Notes
Leads List	/leads	Refactor for M3	Replace hardcoded columns with metadata-driven layout. Saved views from M3. Bulk actions, filters, sources, score range.
Lead Detail	/leads/:id	Refactor for M3	Header + Tabs: Overview (layout-driven from M3) · Timeline · Activities · Related Deals · Files · Audit. Quick actions: Call, Message, Email, Schedule, Convert to Deal.
Create Lead	/leads/create	Refactor for M3	Form rendered from M3 metadata for the user's role-specific create layout.
Lead Kanban	/leads/kanban	Reuse + M3 fields	Drag-drop between stages. Card displays role-specific summary fields from M3 layout.
Lead Scoring	/leads/scoring	Reuse	Rule-based + AI score config and review.
Duplicate Management	/leads/duplicates	Reuse	Strict/soft/fuzzy match queue. Merge with field-level conflict resolution.
CSV Import Wizard	/leads/import	NEW	Step 1 Upload → Step 2 Field Mapping (auto-suggest with manual override, custom fields from M3) → Step 3 Dedup Strategy → Step 4 Preview (first 20 rows) → Step 5 Commit (async, progress) → Step 6 Result (success/error report download). Bulk undo within 24h.
Lead Assignment Config	/leads/assignment	Reuse	Round-robin / rule-based / AI-assisted. Generalize rules to use M3 fields.
Lead Form Configuration (legacy)	/settings/lead-form	Deprecate	Replace with M3 Layout Editor. Legacy redirect.
Webhook Source Configuration	/leads/sources	NEW	List of inbound webhook sources (Web Form, Meta Lead Ads, LinkedIn, IndiaMART, Shopify, WooCommerce, Generic Webhook).

			Each with status, lead count today, error count.
Source Configuration Detail	/leads/sources/:source	NEW	Per-source: connection status, field mapping (source field → M3 lead field), test trigger, recent payloads viewer, error log.
Bulk Operations Toolbar	(component)	NEW	Available on Leads List when rows selected: Assign, Tag, Update Field, Delete, Add to Campaign, Export. Confirmation modal with affected count.

Customizations & Special Behavior

- **Schema-driven UI:** every Lead screen reads its visible fields, validation, and layout from M3 store. Adding a custom field in M3 immediately reflects here without code changes.
- **Source attribution:** every lead has Source + Source Metadata (campaign, channel, ref ID) + first-touch timestamp. Immutable once captured. Visible on detail header.
- **Hybrid scoring:** rule-based score + AI score shown side-by-side. Composite formula visible. AI confidence rendered as a smaller indicator below AI score.
- **Deduplication:** real-time on create (toast: 'Possible duplicate found') + offline queue (Duplicates page). Strict match auto-merges, soft match flags, fuzzy queues for human review.
- **Unified timeline:** append-only event stream showing AI calls (with player), WhatsApp threads, SMS, emails, notes, stage changes, campaign touches, AI qualifications. Filter by event type. AI events visually distinguished.
- **Bulk operations:** select all, with affected count, with progress modal, with undo within 24h.

CRUD Operations

Action	Trigger	Mock Behavior
Create Lead	Form Save	POST → lead in store. Dedup check runs. Audit logged. Optional: trigger workflows.
Update Stage	Drag in Kanban OR Edit form	PATCH stage. Workflow hooks fire. Stage history tracked.
Bulk Import	CSV wizard final step	Async simulation: progress bar updates per 1000 rows. Errors collected. Result exportable.
Merge Duplicates	Duplicates page → Merge	Two-record diff with field-level conflict resolution. Survivor record retains all timeline events from both. Audit logged.

Convert Lead to Deal	Detail page → 'Create Deal' button	Routes to /deals/create with lead context pre-filled. Lead remains, links to deal.
Reassign Lead	Detail page → Reassign	Modal with user picker. Audit logged with previous owner, new owner, trigger source.
Add Note / Activity	Detail page → Quick action	Appended to timeline. Mentions trigger notifications (mock).

Navigation Connections

- Sidebar → Leads → list
- Lead row click → detail
- Detail → 'Create Deal' → deals/create with context
- Detail → Activities tab → log call/message/email
- Top of list page: 'Import' button → CSV wizard
- Top of list page: 'Sources' button → webhook source config

Demo Walkthrough

Tenant Admin opens Leads → Import → uploads 100-row CSV → maps columns to fields (including 2 custom fields from M3) → dedup strategy 'Auto-merge on phone' → previews → commits. Watches progress. 95 leads created, 5 duplicates auto-merged. Opens a lead → sees AI call recording in timeline → converts to Deal.

Existing Files (Reuse / Refactor)

src/app/pages/customer/LeadManagement.tsx · LeadDetail.tsx · LeadKanban.tsx · LeadScoringDashboard.tsx · DuplicateLeadManagement.tsx · LeadGeneration.tsx · components/leads/ · hooks/useLeads.ts (481 L)*

Module 5 — Deal, Opportunity & Pipeline Management

Mission: Revenue tracking. Multiple pipelines, line items, discount approvals, forecasting, AI advisory, attribution, mandatory close reasons.

Pages to Build / Refactor

Page	Path	Status	Notes
Deal Dashboard	/deals/dashboard	Refactor	KPI tiles: open value, won this month, weighted forecast, average velocity. Pipeline charts.
Deals List	/deals/list, /deals (canonicalize)	Refactor	Table view + Kanban toggle. Filters by pipeline, owner, stage, value range.
Deal Pipeline (Kanban)	/deals/pipeline	Reuse	Drag-drop. Stage WIP limits visible if configured.
Create Deal	/deals/create	Reuse + Approval	Already 1066 lines. Add discount-threshold approval branching.
Deal Detail	/deals/:id	Refactor	Tabs: Overview · Line Items · Activity · Related · Forecast · Attribution · Audit.
Deal Edit	/deals/:id/edit	NEW (currently Placeholder)	Edit form. Some fields lock after Won.
Deal Forecasting	/deals/forecasting	NEW (currently Placeholder)	Stage-weighted forecast by period. Owner / pipeline filters. Manual probability override per deal. Time-based projections (this month / quarter / next quarter). CSV/PDF export.
Deal Progression Report	/deals/progression	Reuse	
Discount Approval Queue	/deals/approvals	NEW	Pending discount approvals across deals. Approver actions: Approve, Reject (with comment). Routes back to deal on action.
AI Insights Panel	(side panel on Deal detail)	NEW	Health Score gauge, Risk Flags list, Next-Best-Action suggestions. Each action has 'Execute' (with confirmation) or 'Dismiss' (with optional reason for AI learning).
Attribution Tab	(tab on Deal detail)	NEW	Model selector (First-touch / Last-touch / Linear / Time-decay /

			U-shaped). Contributing campaigns + AI calls + messages with weighted contribution. Drill-through to source.
Close Deal Modal	(modal)	NEW	Mandatory Won/Lost. Mandatory Close Reason (tenant-configurable picker). Final amount lock. Warning that deal will be locked.
Companies & Contacts	/companies, /contacts	NEW (currently Placeholder)	List + detail pages for related entities. M3-driven.

Customizations & Special Behavior

- **Multiple pipelines:** tenant-defined pipelines (Sales, Renewals, Upsell). Each with its own stages, probabilities, gating rules. Deal switches pipeline = stage history versioned.
- **Line items with approval:** discount % > threshold (config) routes deal to 'Pending Approval' status. Approver receives notification. Approval/rejection logged. Deal returns to active state on approval.
- **AI advisory only:** AI never moves stages or modifies values. UI clearly labels: 'AI Suggestion — your decision'. Explicit Execute / Dismiss buttons.
- **Mandatory close reason:** deal cannot move to Won/Lost without selecting a tenant-defined reason. Reason picker shown on close action.
- **Lock-on-close:** closed deals show value/stage as read-only. 'Reopen' action requires Manager+ permission with logged reason.
- **Forecast transparency:** every projected number has a 'How is this calculated?' tooltip showing formula and inputs.

CRUD Operations

Action	Trigger	Mock Behavior
Create Deal	Wizard final step	POST → deal. Optional: trigger workflows. Lead linked.
Add Line Item	Deal detail → Line Items tab → Add	Auto-compute subtotals. Discount > threshold → status = 'Pending Approval'.
Approve Discount	Approval queue → Approve	Status returns to active. Notification to requester.
Move Stage	Drag in Kanban or Edit form	Stage gating evaluated. Required-on-stage fields enforced.
Close Deal Won	Detail → Close → Won	Modal: pick reason, confirm. Lock. Trigger downstream workflows. Attribution finalized.
Reopen Closed Deal	Closed deal → Reopen	Permission-gated. Reason mandatory. Audit logged.

Navigation Connections

- Sidebar → Deals → list
- Deal row click → detail
- Detail → Lead link → Lead detail
- Detail → Activities → trigger AI call / message
- Forecast page filterable by pipeline + owner + period
- Approval queue badge on sidebar shows pending count

Demo Walkthrough

Sales Agent creates deal from lead. Adds 3 line items. Applies 25% discount → triggers approval. Manager reviews approval queue → approves. Agent moves deal through stages. AI Insights flag stalled deal at 'Negotiation' for 14 days → Agent acts on Next-Best-Action. Closes Won with reason 'Reference customer'. Attribution shows 60% contribution from 'Q1 Outreach' campaign.

Existing Files (Reuse / Refactor)

src/app/pages/customer/CreateDeal.tsx (1066 L) · DealDashboard.tsx · DealList.tsx · DealPipeline.tsx · DealDetail.tsx · DealProgressionReport.tsx · hooks/useDeals.ts (582 L)

Module 6 — AI Calling Engine

Mission: The platform's commercial differentiator. Outbound + inbound AI calls, agent config, conversation flow design, live monitor, transfer queue, cost safeguards, consent.

Pages to Build / Refactor

Page	Path	Status	Notes
AI Calling Dashboard	/ai-calling	Reuse	KPIs: today's calls, containment rate, average duration, cost per minute. Trends.
Live Monitor	/ai-calling/live-monitor	Reuse	Real-time call list (mock WebSocket). Click any call → live transcript drawer.
Call Logs	/ai-calling/call-logs	Reuse	Searchable + filterable call history.
Call Detail	/ai-calling/call-detail/:id	Refactor	Audio player + transcript + AI actions taken + outcome + cost. Add: explicit state machine timeline.
AI Agents List	/ai-calling/ai-agents	Reuse	All tenant agents. Tenant-scoped (NOT user-scoped) badge prominent.
Edit Agent	/ai-calling/agents/:id/edit	Reuse + polish	Type, voice, persona, languages, default objectives, forbidden topics.
Conversation Flow Editor	/ai-calling/conversation-flow-editor	Reuse + polish	Already 754 lines. Add: real-time validation indicator, deterministic-replay toggle, version history.
Prompt Template Editor	/ai-calling/prompt-template-editor	Reuse	
Inbound Call Setup	/ai-calling/inbound-call-setup	Reuse	Routing rules, IVR alternative, human fallback. M13 number assignment shown.
Transfer Queue	/ai-calling/transfer-queue	Reuse	Real-time queue. Take Call action by available agent.
Call Analytics	/ai-calling/call-analytics	Reuse	
Call Library	/ai-calling/call-library	Reuse	
Call Scheduling & Retry	/ai-calling/call-scheduling-retry	Reuse	Per-campaign retry rules. Show: max attempts, time between, time windows, channel fallback.

Cost Governance	/ai-calling/cost-governance	Reuse + polish	Add per-campaign max-duration, silence-timeout, loop-detection explicit cards.
Compliance & Consent	/ai-calling/compliance-consent	Reuse + runtime	Tenant config + per-call consent log. Add: consent capture flow visualization.
Cost Transparency Report	/ai-calling/cost-transparency	Reuse	
Live Consent Capture (in-call)	(component)	NEW	Shown when active call requires consent. AI delivers disclosure → records affirmative response → logs with timestamp.
Call State Machine Visualization	(component on detail)	NEW	Shows: Initialized → Dialing → Connected → Active → Action Execution → Wrap-up → Completed. Each state with timestamp.

Customizations & Special Behavior

- **Tenant-scoped agents:** every agent badge says 'Tenant Agent' (not user). Agent attribution shown as: AI Agent ID + Campaign ID + Tenant ID.
- **Two flow design modes:** prompt-based (text editor for fast setup) and flow-based (node editor for regulated workflows).
- **Strict state machine:** every call shows its state. Cannot skip states. Failed states have reason + recovery action visible.
- **Consent disclosure flow:** before recording starts, AI prompts disclosure. Affirmative response required. UI shows consent timestamp on every recorded call.
- **Cost safeguards visible:** max-duration, silence-timeout, loop-detection — each as a separate card with current/configured values. Force-termination event clearly logged.
- **Warm transfer w/ context:** when transferred, human agent sees: AI summary, transcript so far, captured fields, suggested response. Billing-stop timestamp shown.
- **Failure handling visible:** ASR failure / LLM failure / tool failure / silence timeout — each fallback action visible in call detail.
- **Multi-language support:** agent config has supported languages list. Mid-call language switch visible in transcript with timestamp.

CRUD Operations

Action	Trigger	Mock Behavior
Create Agent	Agents → New	POST → tenant-scoped agent. Agent ID generated. Audit logged.

Schedule Outbound Call	Lead detail or Campaign	Mock: call enters queue. Live monitor shows it as Initialized → Dialing.
Take Inbound Call	Mock inbound trigger	Routing rules evaluated. Number → tenant. AI agent loaded. State: Connected.
Capture Consent	(during call)	Mock: AI delivers disclosure. UI shows consent prompt. Affirmative recorded. Recording starts.
Warm Transfer	Live monitor → Transfer	Human agent picker. Selected agent receives notification. Context summary shown to agent. AI billing stops, human billing starts.
Force Terminate	Live monitor → Force End (admin only)	Confirmation modal with reason. Call ends. Logged.
Save Conversation Flow	Flow editor → Save	New version. Existing in-flight calls continue on previous version.

Navigation Connections

- Sidebar → AI Calling section (10 items)
- Live Monitor → click call → side drawer with live transcript
- Call Detail → 'View Lead' → Lead detail
- Agent Edit → 'Edit Conversation Flow' → Flow Editor
- Transfer Queue → 'Take Call' → live transcript joins
- Compliance page → per-call consent log links to call detail

Demo Walkthrough

Tenant Admin creates AI Agent 'Sarah-Sales' with Hindi + English. Designs flow w/ Speak → Listen → Decision → Action → End nodes. Schedules outbound campaign. Live Monitor shows 5 calls in flight. One call: customer requests human → warm transfer → human agent takes over with context summary. Another: silence timeout → AI ends call gracefully. All calls visible in Call Logs with full transcripts.

Existing Files (Reuse / Refactor)

src/app/pages/customer/AICallingDashboard.tsx · LiveMonitor.tsx (648 L) · CallLogs.tsx (785 L) · CallDetail.tsx (641 L) · ConversationFlowEditor.tsx (754 L) · EditAgent.tsx (962 L) · ComplianceConsent.tsx (827 L) · CostGovernance.tsx (752 L) · TransferQueue.tsx · hooks/useAICalling.tsx (231 L)

Module 7 — WhatsApp CRM

Mission: Compliant AI+human hybrid conversation channel. Meta API mediated by platform. Templates approval lifecycle, broadcast caps, opt-out enforcement.

Pages to Build / Refactor

Page	Path	Status	Notes
WhatsApp Dashboard	/whatsapp	Reuse	
WhatsApp Setup	/whatsapp/setup	Reuse	Platform-mediated Meta connection. Tenant verifies via Meta separately.
WhatsApp Inbox	/whatsapp/inbox	Reuse + polish	AI/Human ownership labels prominent. Take Over / Hand Back to AI buttons clear.
WhatsApp Leads	/whatsapp/leads	Reuse	
Templates List	/whatsapp/templates	Reuse + Meta states	Status badges: Draft, Submitted, Approved, Rejected (w/ Meta reason).
Create / Edit Template	/whatsapp/templates/create, /:id/edit	Reuse + polish	Variable picker, CTA buttons (call/URL), preview.
Campaigns List	/whatsapp/campaigns	Reuse	
Create Campaign	/whatsapp/campaigns/create	Reuse	Already 1053 lines.
Automation Rules	/whatsapp/automation	Reuse	Inbound triggers + auto-replies.
Create Automation	/whatsapp/automation/create	Reuse	
Consent Management	/whatsapp/consent	Reuse	Per-recipient consent state. Single/double opt-in tracking. Revocation history.
WhatsApp Analytics	/whatsapp/analytics	Reuse	
WhatsApp API	/whatsapp/api	Reuse + framing	Make clear: tenants do NOT supply Meta credentials. Platform-mediated.
WhatsApp Settings	/whatsapp/settings	Reuse	
Take-Over / Hand-Back UX	(in inbox)	NEW	Explicit buttons in active conversation header. SLA timer reset on transition. Both directions logged.

Customizations & Special Behavior

- **Meta template approval lifecycle:** Draft → Submitted → (Approved | Rejected). Rejected templates show Meta's reason. Approved templates can be edited (creates new version requiring re-approval).
- **Session window enforcement:** outside 24-hour window, free-text composer is disabled, only approved-templates picker available. Banner shows: 'Outside session window — template-only.'
- **AI ↔ Human ownership:** every conversation has an Ownership badge: AI / Human / Hybrid. Take Over button visible to permitted roles. Hand Back available.
- **Opt-out enforcement:** STOP keyword adds to opt-out list. Future sends to that number show 'Blocked — Opted Out' indicator. Cannot be overridden.
- **Broadcast caps:** campaign settings show daily cap, rate limit. When cap reached, queue paused (visible). When opt-outs occur mid-campaign, count visible.
- **Media handling:** images/audio/documents shown inline in conversation. File-size limits enforced.

CRUD Operations

Action	Trigger	Mock Behavior
Receive Inbound Message	Mock trigger every 30s	Auto-create lead if unknown OR append to existing thread. AI replies if AI-owned. Toast notification.
Send Template Message	Composer outside session	Template picker only. Variables filled. Status: Sent → Delivered → Read.
Submit Template for Approval	Templates → Submit	Status → Submitted. Mock 3-second wait → Approved or Rejected randomly.
Run Broadcast Campaign	Campaigns → New	Audience filter (M4 Lead segments). Schedule. Caps. Opt-out filter applied. Execution monitor live.
Take Over Conversation	Inbox → Take Over	AI replies pause. Composer enabled for human. Audit logged.
Receive STOP	Mock	Opt-out registered. Confirmation reply sent (per regulation). All future sends blocked.
Hand Back to AI	Inbox → Hand Back	Ownership transitions. AI receives summary of human exchange. SLA reset.

Navigation Connections

- Sidebar → WhatsApp section
- Inbox → conversation → 'View Lead' → Lead detail
- Lead detail → 'Send WhatsApp' → opens composer with thread
- Templates list → row → preview drawer
- Campaigns → row → execution monitor

- Consent page → per-recipient → opt-out history

Demo Walkthrough

Tenant Admin submits a 'Order Confirmation' template → Meta approves → activates broadcast campaign to 1000 leads. Watches caps trigger pause at daily limit. Reviews inbox: 47 replies, 30 handled by AI, 5 escalated to human. Manager takes over a complex one. Customer replies STOP — opt-out enforced.

Existing Files (Reuse / Refactor)

12 dedicated pages: src/app/pages/customer/WhatsApp.tsx · CreateWhatsAppCampaignAdvanced.tsx (1053 L) · components/whatsapp/* · hooks/useWhatsApp.ts (447 L) + useWhatsAppAPI.ts (335 L)*

Module 8 — SMS CRM

Mission: High-deliverability transactional + promotional SMS. India DLT compliance. Quiet hours, opt-out, keyword-driven workflows.

Pages to Build / Refactor

Page	Path	Status	Notes
SMS Dashboard	/sms	Reuse	
SMS Inbox	/sms/inbox	Reuse	
Campaigns	/sms/campaigns	Reuse	
Create SMS Campaign	/sms/campaigns/create	Refactor	Add: DLT category picker, sender ID picker (DLT-registered only), quiet hours config (TRA1 9pm-9am for promotional).
SMS Templates	/sms/templates	Refactor	DLT registration status badge per template. Cannot send unregistered templates (button disabled with explanation).
Short Links	/sms/short-links	Reuse	
SMS Analytics	/sms/analytics	Reuse	
SMS Settings	/sms/settings	Reuse	
DLT Registry (NEW)	/sms/dlt-registry	NEW — CRITICAL India compliance	Tenant's DLT Entity ID, Sender IDs (Header IDs), Templates registered. Each template: DLT Template ID, Category (Transactional/Promotional/Service-Implicit/Service-Explicit), Status (Pending/Approved/Rejected). Submit Template action.
Submit Template to DLT (modal)	(component)	NEW	Wizard: select template body, pick category, attach DLT Entity, submit. Status moves to Pending → Approved (mock 5s).
Quiet Hours / DND Window Enforcement	(in campaign create)	NEW	Time picker for allowed send window per day. India promotional DLT defaults: 9am-9pm. Validation prevents schedule outside window.
Keyword Routing Configuration	/sms/keywords	NEW	Map inbound keywords to actions: 'INTEREST' → create lead in pipeline X · 'STOP' → opt-out · 'YES' → confirm appointment · custom.

Customizations & Special Behavior

- **DLT compliance is non-negotiable for India:** every sender ID and template must show DLT registration status. Send button disabled if template not DLT-approved. Toast on attempt: 'Cannot send — template not registered with TRAI DLT.'
- **Category-driven routing:** Transactional templates can send 24/7 to any opted-in number. Promotional templates respect TRAI 9 PM – 9 AM blackout. Service-Explicit follows opt-in rules.
- **Opt-out keyword auto-detection:** STOP, UNSUBSCRIBE, STOPALL, CANCEL — all auto-trigger opt-out. Confirmation reply sent automatically (regulator requirement). Future sends blocked.
- **Quiet hours enforcement:** campaigns scheduled to send during DND window are queued until window opens. Visible: 'Send paused until 9 AM tomorrow'.
- **Transactional vs Promotional vs Service-Implicit:** campaign create wizard forces category selection. UI visually distinguishes (warmer colors for promotional, neutral for transactional).

CRUD Operations

Action	Trigger	Mock Behavior
Register DLT Template	DLT Registry → Submit Template	Mock: status → Pending → Approved (5s). Approved badge. Campaign send button enabled.
Create Campaign	Campaigns → New	Wizard: pick template (DLT-approved only), audience, schedule (respects quiet hours), caps. Save.
Run Campaign	Campaign detail → Activate	Execution monitor live. Sends pause at daily cap. Resume next day.
Receive Inbound SMS	Mock trigger	Keyword routing evaluated. STOP → opt-out + confirmation reply. INTEREST → lead created.
Inbox Reply	Inbox → reply to thread	Free-text in active conversation. AI suggests draft.
Configure Keyword	Keyword Routing → Add	Action picker: Create Lead / Opt-Out / Trigger Workflow / Schedule Callback. Save.

Navigation Connections

- Sidebar → SMS section
- DLT Registry on first visit shows setup wizard if no entity registered
- Templates list → 'Register with DLT' for unregistered templates
- Inbox → conversation → 'View Lead'

Demo Walkthrough

Tenant Admin sets up DLT Entity ID. Submits 5 templates → 4 approved, 1 rejected. Creates promotional campaign for approved 'New Year Offer' template. Schedules send for 5 PM IST. Confirms opt-outs registered. Campaign sends 800 SMS, opt-outs received from 12, leads created from 6 'INTEREST' replies.

Existing Files (Reuse / Refactor)

src/app/pages/customer/SMS.tsx · CreateSMSCampaign.tsx (740 L). DLT registry — BUILD FROM SCRATCH.*

Module 9 — Email CRM

Mission: Long-form, high-context channel. Inbound auto-create-lead, AI drafting, drip sequences, deliverability monitoring, suppression list.

Pages to Build / Refactor

Page	Path	Status	Notes
Email Dashboard	/email	Reuse	
Email Inbox	/email/inbox	Reuse	781 lines. Polish AI authorship distinction.
Email Compose	/email/compose	Reuse + AI Draft	Add 'AI Draft' button — generates draft from lead context, last call, last message. User edits before send.
Email Campaigns	/email/campaigns	Reuse	
Create Email Campaign	/email/campaigns/create	Reuse + drip	Multi-step drip sequence builder with delay nodes, exit criteria (reply / unsubscribe / stage change).
Email Campaign Edit	/email/campaigns/:id/edit	Reuse	
Email Templates	/email/templates	Reuse	Approval lifecycle: Draft → Approved.
Create / Edit Template	/email/templates/create, /:id/edit	Reuse	
Email Signatures	/email/signatures	Reuse	Tenant-level, role-based, user-specific.
Email Analytics	/email/analytics	Reuse	
Email Settings	/email/settings	Reuse	Sender addresses, SMTP-platform info.
Deliverability Dashboard	/email/deliverability	NEW	SPF / DKIM / DMARC alignment per sender domain. Bounce rate, complaint rate, reputation score gauges. Alerts when thresholds crossed (5% bounce / 0.3% complaint).
Suppression List	/email/suppression	NEW	Global suppression: hard bounces, unsubscribes, spam complaints. Each entry: reason, date, source. Cannot remove without dual approval.

Unsubscribe Confirmation Page	/u/:token (public)	NEW	Public landing page when recipient clicks unsubscribe link in email. One-click confirm.
-------------------------------	--------------------	-----	---

Customizations & Special Behavior

- **AI draft mode:** compose page has AI Draft button. Generated drafts respect tenant brand voice + signature. User edits remain after subsequent suggestions.
- **Drip exit criteria:** drip sequences have explicit exit conditions: reply received / unsubscribed / stage changed / manual stop. Visible per drip stage.
- **Suppression enforcement:** send to suppressed address shows pre-flight warning: 'X recipients are on suppression list and will be excluded.' Cannot be overridden.
- **Deliverability dashboard:** SPF/DKIM/DMARC live status per sender domain. Bounce rate / complaint rate trend lines. Reputation score 0–100. Auto-throttle indicator if reputation drops.
- **Send window:** drip steps respect send window (e.g., 10am-6pm local). Sends outside queue until window opens.
- **Attachment scanning:** every attachment shows 'Scanned' badge after mock 2-second scan. Malware detected → blocked + alert.

CRUD Operations

Action	Trigger	Mock Behavior
AI Draft Email	Compose → AI Draft	Mock 2-second loading. Draft populated. User edits and sends.
Send Email	Compose → Send	Pre-flight: suppression check, rate limit check. Toast on send. Status: Sent → Delivered → Opened → Clicked tracked.
Run Drip Sequence	Campaign Activate	Leads enter drip. Each step fires per delay. Exit on reply/unsubscribe/stage.
Process Inbound Email	Mock trigger	Auto-create lead if unknown. Append to thread if known. AI suggests reply.
Add to Suppression	Hard bounce / unsubscribe / complaint	Auto-add. Visible in suppression list. Future sends blocked.
Remove from Suppression	Suppression → Remove	Maker-checker required. Reason mandatory. Audit logged.

Navigation Connections

- Sidebar → Email section (10 items)
- Inbox → message → 'View Lead'
- Lead detail → 'Send Email' → opens composer with thread context
- Campaign → row → execution monitor

Demo Walkthrough

Tenant Admin schedules a 4-step drip campaign. AI drafts each step from template. User reviews and approves. Campaign runs. Lead replies → exits drip. Another lead unsubscribes → added to suppression. Deliverability dashboard shows bounce rate creeping to 4% → alert.

Existing Files (Reuse / Refactor)

src/app/pages/customer/Email.tsx · EmailInbox.tsx (781 L) · EmailSettings.tsx (724 L) ·
hooks/useEmail.ts (574 L)*

Module 10 — Campaigns & Workflow Automation (Node-Based)

Mission: The orchestration brain. Node-based workflow builder for multi-channel sequences. A/B testing, budget caps, dynamic segmentation, deterministic execution.

Pages to Build / Refactor

Page	Path	Status	Notes
Campaign Manager (List)	/campaigns, /campaign-manager	Reuse	
Campaign Dashboard	/campaigns/dashboard	Reuse	KPI tiles: active campaigns, today's spend, conversions, ROI.
Campaign Create	/campaigns/create	Reuse	Already 776 lines. Connects to workflow builder for orchestration.
Create Omni Campaign	/campaigns/create-omni	Reuse	Already 883 lines. Multi-channel quickstart wizard.
Campaign Detail	/campaigns/:id	NEW (currently Placeholder)	Tabs: Overview · Workflow (visual) · Execution Log · Audience · Budget · Analytics · Audit. Pause/Resume/Stop with confirmation.
Campaign Edit	/campaigns/:id/edit	NEW (currently Placeholder)	
Campaign Analytics	/campaigns/:id/analytics	NEW (currently Placeholder)	Funnel, channel breakdown, A/B winner, ROI.
Campaign Budget Manager	/campaigns/:id/budget	Reuse	Total / Daily / Per-Lead caps. Burn rate gauge. Auto-pause indicator.
Campaign Execution Monitor	/campaigns/:id/monitor	Reuse	Real-time leads in flight per step. Stuck leads alert.
Workflow Builder (Canvas)	/workflow, /workflow-automation	Reuse + polish	Already substantial (~976 L across 4 components). Polish: real-time validation, version history, deterministic-replay simulation.
Workflow List	/workflows	NEW	All saved workflows. Active/paused/archived states.
A/B Test Configuration	(within campaign)	NEW	Setup variant A vs B with traffic split (default 50/50). Confidence

			threshold for winner promotion. Cost preview before activate.
A/B Test Results	/campaigns/:id/ab-test	NEW	Live performance per variant. Statistical significance indicator. 'Promote A' / 'Promote B' actions.
Audience Builder	(modal in campaign)	NEW	Filter rules (M3-driven for custom fields). Static (frozen at start) vs Dynamic (re-evaluated per step) toggle. Audience size preview before save.
Workflow Version History	(side panel)	NEW	Last 30 versions of any workflow. Diff viewer. Active version + draft indicator. In-flight leads complete on version they started.

Customizations & Special Behavior

- **Real node-based builder:** node palette (Triggers / Actions / Decisions / Delays / Exits) on left, drag onto canvas, connect with edges, configure each node in right inspector. Validation prevents disconnected/cyclic graphs.
- **Node types comprehensive:** Triggers: Lead Created, Stage Change, Time Schedule, AI Outcome, Inactivity, Webhook · Actions: AI Call, WhatsApp, SMS, Email, Task Create, Field Update, Lead Update · Decisions: Call Outcome, Score, Field Condition, Time Window · Delays: Fixed, Business-Hour Aware, SLA-Aware · Exit: Tag with Reason.
- **Audience segmentation:** filter builder uses M3 metadata. Custom fields appear automatically. AND/OR groups. Live audience size shown.
- **Budget enforcement:** real-time burn rate + cap gauges. When cap reached, campaign auto-pauses with notification. Resume requires manual action.
- **A/B testing:** single test per campaign. Traffic split. Cost impact preview. Statistical confidence visualizer. Auto-promote winner option.
- **Pause vs Stop semantics:** Pause = no new steps, in-flight actions complete (live AI calls finish). Stop = cancel all queued + tag exited leads with reason.
- **Workflow versioning:** edits to active workflow create new version. In-flight leads continue on previous version. New leads use new version. Visible in version history.

CRUD Operations

Action	Trigger	Mock Behavior
Build Workflow	Workflow Builder canvas Save	POST → workflow saved with version. Validation errors prevent save.
Activate Campaign	Campaign Detail → Activate	Audience filter applied. Triggers begin evaluating. Execution monitor live.

Pause Campaign	Campaign Detail → Pause	New steps stop within ~10s. In-flight actions continue.
Stop Campaign	Campaign Detail → Stop	Confirmation: 'X leads will exit with reason Campaign-Stopped'. On confirm, queue cancelled.
Setup A/B Test	Campaign → New A/B Test	Cost preview shown before activation. On activate, traffic split begins.
Promote Winner	A/B Results → Promote A	Remaining traffic routed to A. Test concluded.
Edit Active Workflow	Builder → Save while running	New version. Modal: 'In-flight leads will complete on previous version'.

Navigation Connections

- Sidebar → Campaigns & Workflows section
- Campaign Detail → 'Workflow' tab → in-page Builder canvas
- Workflow Builder → 'Save & Activate' → routes back to campaign
- Execution Monitor → click stuck lead → Lead detail with current step highlighted
- A/B Results → click variant → variant message preview

Demo Walkthrough

Tenant Admin opens Workflow Builder. Drags nodes: Trigger (New Lead) → AI Call → Decision (No Answer?) → WhatsApp Template → Decision (Replied?) → Create Deal. Saves. Activates campaign with audience 'Score > 70 AND Source = Meta'. Sets budget: ₹50,000 total. Watches Execution Monitor: 200 leads in flight. Sets up A/B test on the WhatsApp message variant. After 4 hours, Variant B winning by 28% — promotes. Budget hits 80% → soft alert. Pauses campaign for review.

Existing Files (Reuse / Refactor)

src/app/pages/customer/CampaignManager.tsx (687 L) · CreateCampaign.tsx (776 L) · CreateOmniCampaign.tsx (883 L) · WorkflowAutomation.tsx (334 L) · components/workflow/ (976 L total: Canvas, Node, Sidebar, ConfigPanel) · CampaignBudgetManager.tsx · CampaignExecutionMonitor.tsx · hooks/useCampaigns.ts (640 L)*

Module 11 — Business Playground & AI Lead Generation

Mission: Growth tier. AI research, knowledge base, content creation, social media, lead discovery via scraping. Credit-metered with pre-execute cost preview.

Pages to Build / Refactor

Page	Path	Status	Notes
Playground Home	/playground	Reuse	Cards for Research, KB, Assets, Content, Social, Lead Gen, Strategy, Responses.
AI Research	/playground/research, /research	Reuse	Domain-bounded sessions with citations. Saved research.
Knowledge Base	/playground/knowledge, /knowledge	Reuse + role access	326 lines. Add per-document role access (M2 integration).
Asset Library	/playground/assets, /assets	Reuse	
Content Creation Studio	/playground/content, /content	Reuse	Brand voice config, tone, forbidden terms.
Social Media Marketing	/playground/social, /social	Reuse	Connected accounts (LinkedIn, X, Facebook, Instagram). Schedule, post.
Lead Generation (Scraping)	/playground/leads-generation, /leads-generation	Reuse + allowlist	Scraping criteria. Add: visible allowlist + robots.txt status per source.
Strategy Planning	/playground/strategy, /strategy	Reuse	
Response Management	/playground/responses, /responses	Reuse	
Storage Manager	/storage	Reuse	
Credit Cost Preview Modal	(component, used everywhere)	NEW	Universal pattern: before any AI / scraping / credit-consuming action, modal shows: 'This will consume X credits. Current balance: Y. Confirm?' Buttons: Confirm / Cancel.
OAuth Connection Manager	/playground/connections	NEW	Per-user social account connections. Status, last-refreshed, disconnect button.

Customizations & Special Behavior

- **Universal credit preview:** EVERY action that consumes credits shows pre-execute modal with exact cost. Cancel returns no debit. Partial completion bills only successful portion.
- **Citation requirement:** AI research outputs include inline citations with URLs. Broken or hallucinated citations flagged on output review.
- **KB tenant-isolated:** uploaded documents are tenant-scoped (cannot leak to other tenants). Per-doc role access controls visibility within tenant.
- **Brand voice config:** tenant-level: tone descriptors, forbidden words, preferred phrases, target audience. Generated content respects these.
- **Robots.txt + allowlist:** scraping UI shows which sources are allowed (whitelist + robots.txt-respecting). Attempts to scrape blocked sources show informative error.
- **OAuth tokens user-scoped:** each user connects their own social accounts. Tokens never shared at tenant level (privacy).

CRUD Operations

Action	Trigger	Mock Behavior
Run Research Session	Research → New Session	Cost preview modal (e.g., 3 credits). Confirm. Mock 4-second loading. Output with citations.
Upload KB Doc	KB → Upload	Mock 2-second indexing. Searchable + queryable in KB queries.
Generate Content	Content → New	Cost preview. Brand voice applied. Generated draft shown with regenerate / edit / push-to-templates actions.
Scrape Leads	Lead Gen → New	Criteria + cost preview + allowlist check. Mock results 100 leads. Dedup against existing → 87 new. Bulk accept to CRM.
Schedule Social Post	Social → Compose → Schedule	OAuth check. Schedule. Visible in calendar.
Save Research	Research → Save	Saved under user. Retrievable in saved-research list.

Navigation Connections

- Sidebar → Business Playground section
- Each playground module → 'Push to CRM' / 'Push to Templates' actions to integrate with main CRM
- Lead Gen results → bulk select → push to /leads
- Content draft → push to /email/templates or /whatsapp/templates

Demo Walkthrough

Tenant Admin opens Research → 'Top 10 EV charging companies in Bangalore' → confirms 3 credits → reviews output. Saves. Opens Lead Gen → similar criteria → confirms 15 credits →

100 leads scraped → bulk imports 87 new ones to /leads. Generates a WhatsApp template from Content Studio → pushes to /whatsapp/templates draft.

Existing Files (Reuse / Refactor)

src/app/pages/customer/BusinessPlayground.tsx · ProductResearch.tsx · KnowledgeBase.tsx (326 L) · LeadGeneration.tsx · SocialMediaMarketing.tsx · StrategyPlanning.tsx · components/playground/ · components/leads/ScrapingRequestForm.tsx*

Module 12 — Analytics & Reporting

Mission: Role-based dashboards, custom dashboard builder (drag-drop tiles), custom report builder (M3-driven), anomaly insights, scheduled reports.

Pages to Build / Refactor

Page	Path	Status	Notes
Reports & Analytics Hub	/reports-analytics	Reuse	Top-level navigation hub.
AI Calling Report	/reports-analytics/ai-calling	Reuse	
CRM Report	/reports-analytics/crm	Reuse	
Campaigns Report	/reports-analytics/campaigns	Reuse	
Deals Report	/reports-analytics/deals	Reuse	
Performance Report	/reports-analytics/performance	Reuse	
Cost Report	/reports-analytics/cost	Reuse	
Custom Report Builder	/reports-analytics/builder	Refactor + M3	528 lines existing. Add: scheduling, recipient-level redaction, custom-object support from M3.
Custom Dashboard Builder	/dashboards/builder	NEW — CRITICAL	Drag-drop tile composer. Tile types: KPI, line chart, bar, pie, funnel, table, gauge, heatmap. Per-tile data source picker (entity from M3, measure, dimension, filter). Per-role sharing.
My Dashboards	/dashboards	NEW	List of dashboards: predefined (role-based) + custom. Star to pin. Set as default.
Dashboard View	/dashboards/:id	NEW	Render dashboard. Date range picker. Drill-down on any tile to underlying records.
Anomaly Insights Center	/insights	NEW	Cards for anomalies detected (sudden conversion drop, spend spike, channel failure). Severity badges. Dismiss / Act actions.
Scheduled Reports	/reports-analytics/schedules	NEW	List of recurring reports. Each: name, recipients, schedule (daily/weekly/monthly), next run, last run, status.

Tile Configuration Modal	(component)	NEW	Wizard: pick entity (any from M3) → pick visualization → measure (sum, count, avg) → dimension (group by) → filter → preview → save.
--------------------------	-------------	-----	--

Customizations & Special Behavior

- **Drag-drop dashboard builder:** blank canvas. Drag tiles from palette. Resize. Reorder. Per-tile config opens inspector.
- **M3-driven sources:** data source picker reads object list from M3 store. Custom objects appear automatically.
- **Role-aware redaction:** when scheduled report is sent, recipient sees only fields permitted by their role (M2 integration). Hidden fields show 'Redacted'.
- **Anomaly insights:** automatically generated based on metric thresholds (mock). Each card explains: what changed, likely cause, suggested action. Dismissible.
- **Drill-through:** any KPI / chart click navigates to underlying records (e.g., 'Deals Won' → Deal list filtered to Won). Maintains filters.
- **Snapshot scheduling:** scheduled report snapshots data at run-time. Even if records change later, the snapshot is preserved.

CRUD Operations

Action	Trigger	Mock Behavior
Build Dashboard	Dashboard Builder Save	POST → dashboard saved. Sharing config applied.
Build Report	Report Builder Save	POST → report saved. Schedule optional.
Schedule Report	Report → Schedule	Modal: cadence, recipients (filtered to permitted roles only), format (PDF/CSV/XLSX). Save.
Drill-Through	Click KPI on dashboard	Navigates to filtered list page with dashboard filters applied.
Dismiss Anomaly	Insight card → Dismiss	Hidden for current user. Preserved for other admins.
Act on Anomaly	Insight card → Suggested action	Routes to relevant page with context (e.g., 'Adjust campaign budget' → Budget Manager).

Navigation Connections

- Sidebar → Reports & Analytics
- Each predefined report has 'Customize' button → opens Report Builder with that report pre-loaded
- Dashboard tile → drill-through to underlying records

- Anomaly insight → 'Suggested Action' → contextual page

Demo Walkthrough

Manager opens Dashboards → New → drags 6 tiles (KPIs + 2 charts + 1 table) → configures each (data source = Deals, measure = Total Value, dimension = Stage, filter = This Quarter) → saves and pins. Schedules a weekly summary report to leadership. Reviews Anomaly Insights — sees 'AI call success rate dropped 32% this week' → drills into call logs → identifies pattern.

Existing Files (Reuse / Refactor)

src/app/pages/customer/ReportsAnalyticsDashboard.tsx · CustomReportBuilder.tsx (528 L) + role reports. Dashboard builder — BUILD FROM SCRATCH.

Module 13 — Integrations — Telephony, Calendar, Webhooks

Mission: Platform-mediated external system integration. Calendar sync, Meta Lead Ads, number inventory, webhook config with HMAC, accounting export.

Pages to Build / Refactor

Page	Path	Status	Notes
Integrations Hub	/integrations	NEW (currently scattered)	Cards: Telephony · Calendar · Lead Sources · Webhooks · Accounting · Communication Providers (display-only). Status per integration.
Telephony — Number Inventory (Super Admin)	/admin/telephony/inventory	NEW	All platform-owned numbers. Filters: country, region, type (toll-free/local/short-code), status (available/assigned/released).
Telephony — Provider Management (Super Admin)	/admin/telephony/providers	NEW	List of telephony providers. Add provider, configure routing, test calls, promote to Active. Quality SLAs displayed.
Telephony — Provider Detail	/admin/telephony/providers/:id	NEW	Per-provider: rate card, regions supported, current quality metrics, test-call action, activate/deactivate.
Number Assignment to Tenant	(action on tenant detail)	NEW	Modal: pick number from inventory + purpose (outbound/inbound/campaign). Assign. Visible in tenant detail.
Calendar Connection	/integrations/calendar	NEW	Connect Google Calendar / Outlook via OAuth. Per-user. Status, last refresh, disconnect.
Calendar Connection Detail	/integrations/calendar/:provider	NEW	Connected account info, sync settings (working hours, free/busy visibility), test event creation.
Outbound Webhooks	/integrations/webhooks	Refactor	List of webhook subscriptions. Per-row: URL, event types, status, last delivery, retry count.
Webhook Detail / Edit	/integrations/webhooks/:id	NEW	Per-subscription: URL, secret (rotate), event types, delivery log (last 30 days), test send button.
Inbound Lead Sources	/integrations/lead-sources	NEW — CRITICAL	Cards per source: Meta Lead Ads, LinkedIn, IndiaMART, Google Ads, Shopify, WooCommerce, Generic

			Webhook. Each: status, leads today, error count, configure.
Lead Source Configuration (per type)	/integrations/lead-sources/:type	NEW	Source-specific OAuth/setup wizard. Field mapping: source field → M3 lead field. Test trigger. Recent payloads viewer.
Accounting Export	/integrations/accounting	NEW	Configure scheduled exports: target (CSV download / SFTP / API push), cadence, filters. Export history.
API Keys (Tenant)	/settings/api	Reuse + polish	Tenant-issued API keys for outbound integration. Scopes, last used, revoke.

Customizations & Special Behavior

- **Platform owns telephony:** tenants do NOT supply provider credentials. UI shows numbers from platform inventory only. 'Bring Your Own Carrier' is explicitly excluded — UI message if requested.
- **Number lifecycle:** Available → Assigned (to tenant) → Released → Cooldown (30d) → Available. Each transition logged.
- **Webhook HMAC + signed payloads:** every outbound webhook signs payload with tenant-secret. Test send button shows actual payload + signature for verification.
- **Retry with backoff:** failed deliveries retry with exponential backoff up to 5 attempts in 24 hours. Visible in delivery log per attempt.
- **Auto-pause failing webhooks:** after 48 hours of consistent failures, subscription auto-pauses. Tenant Admin notified.
- **Calendar OAuth user-scoped:** tokens stored per user. Platform never sees user passwords.
- **Lead source connectors:** Meta Lead Ads (OAuth via Meta Business), LinkedIn Lead Gen (OAuth), IndiaMART (API key), Google Ads (OAuth), Shopify (API key + store URL), WooCommerce (REST API key), Generic Webhook (URL + signature).

CRUD Operations

Action	Trigger	Mock Behavior
Assign Number	Inventory → Assign	Modal: pick tenant + purpose. Status flips to Assigned. Visible in tenant detail.
Connect Calendar	Integrations → Calendar → Connect Google	OAuth simulation. Mock 2-second redirect → return with token. Connected status.
Configure Webhook	Webhooks → New	URL, events, secret. Save. Test send action. Delivery log starts populating.
Connect Lead Source	Lead Sources → Meta Lead Ads → Connect	OAuth simulation. Field mapping wizard. Test trigger creates 1 mock lead. Active.

Receive Inbound Lead via Webhook	Mock trigger every 60s	Source webhook → field-mapped → Lead created → toast 'New lead from Meta Lead Ads'.
Test Send Webhook	Webhook detail → Test Send	Mock POST. Response code, latency, body excerpt shown.
Onboard Provider	Telephony → Providers → Add	Wizard: credentials + regions + rate card + test calls. State: Testing → Active.

Navigation Connections

- Sidebar (Super Admin) → Operations → Telephony Inventory · Provider Onboarding
- Sidebar (Tenant) → Settings → Integrations Hub
- Tenant detail → 'Numbers' section → Assign action
- Lead detail → Source field → click → source configuration
- Webhook detail → 'Resend All Failed' bulk action

Demo Walkthrough

Super Admin onboards new telephony provider 'Provider X' → tests calls → promotes to Active. Purchases 50 numbers. Assigns 10 numbers to tenant. Tenant Admin connects Google Calendar (OAuth). Configures Meta Lead Ads webhook. Receives 5 leads in next hour. Configures outbound webhook to push deal-won events to their CRM. Tests send. Configures weekly accounting export.

Existing Files (Reuse / Refactor)

src/app/pages/admin/settings/APIIntegrations.tsx · pages/customer/StandaloneConfig.tsx (791 L) · InboundCallSetup.tsx (646 L). Most of M13 — BUILD FROM SCRATCH.

Module 14 — Billing, Usage & License Enforcement

Mission: Source of truth for revenue. Per-second metering, GST-compliant invoicing (India), prepaid + offline payment with dual approval, hard-stop propagation.

Pages to Build / Refactor

Page	Path	Status	Notes
Billing Overview (Tenant)	/billing	Reuse	Current plan, balance, recent invoices, soft-alert thresholds.
Usage Metering	/billing/usage-metering, /billing/usage	Reuse + GST	Per-channel usage breakdown. Live ledger view. Display in INR.
Usage Analytics (Tenant)	/billing/usage	Reuse	
Credit Purchase	/billing/purchase	Refactor — INR + GST	Top-up flow with Stripe + Razorpay. INR throughout. GST 18% added on invoice.
Credit Configuration (Super Admin)	/admin/billing/credit-management, /credits	Reuse	
Invoices List	/billing/invoices	Reuse + GSTIN	Tenant's invoices. GSTIN visible per invoice.
Invoice Detail	/billing/invoices/:id	Refactor — full GST	Add: GSTIN of issuer + recipient · HSN/SAC codes per line item · IGST/CGST/SGST breakdown · Tax invoice number (monotonic per FY) · INR amount in words.
Create Invoice (Super Admin)	/admin/billing/invoices/create	Reuse + GST	Generate invoice for tenant. GST auto-calculated based on tenant state vs platform state (CGST+SGST intra-state, IGST inter-state).
Billing Rules	/admin/billing/rules	Reuse	Tax rates, currency, invoice numbering.
Billing History	/admin/billing/history	Reuse	
Subscription (Tenant)	/subscription	Reuse	
Subscription Plans Management (Admin)	/admin/subscription-plans	Reuse — INR	
Payment Methods	/billing/payment-methods	NEW (currently Placeholder)	Saved payment methods (cards, UPI, net banking). Set default. Remove.

Offline Payment Proof Upload	/billing/offline-payment	NEW — CRITICAL	Upload bank transfer / cheque proof. Reference number, amount, date, document. Submits to Finance approval queue.
Offline Payment Approval Queue (Admin)	/admin/billing/payment-s/approval-queue	NEW	Pending offline payment approvals. Per row: tenant, amount, proof. Maker-checker required (2 distinct Finance users).
Billing Alerts Center	/admin/alerts, /alerts	Reuse	Soft alerts at 80%/90%/95% thresholds. Hard stop alerts.
Hard-Stop State Indicator	(component on tenant cards)	NEW	When license expires or credits depleted, tenant card visibly shows 'Hard Stop Active' with badge. AI calls / messages blocked.
Tax Invoice Numbering Display	(everywhere)	NEW	Format: INV-YYYY-XXXXXX (per fiscal year, monotonic). Display in invoice header.
Credit Adjustment (Super Admin)	/admin/billing/credit-adjustment	NEW	Manual credit/debit with reason code + dual approval + linked invoice/credit-note.

Customizations & Special Behavior

- **INR throughout (currency swap):** all NPR references replaced with INR. Subscription plans, invoices, billing dashboards, usage metering. Currency configurable per tenant but defaults to INR for India.
- **GST compliance:** every invoice for India tenant has GSTIN, HSN/SAC codes, IGST/CGST/SGST breakdown. Intra-state: CGST + SGST. Inter-state: IGST. UI auto-applies based on tenant state.
- **Monotonic invoice numbering:** invoice numbers strictly monotonic per tenant per fiscal year (April–March in India). Numbering gaps forbidden. Visible enforcement.
- **Offline payment dual approval:** tenant uploads proof → Finance Approver 1 reviews → routes to Approver 2 → on approval, credits added. Reject reason mandatory if rejected.
- **Hard stop propagation:** when credits hit 0 or license expires, status flips to 'Hard Stop' within 60 seconds (mock). Tenant card visibly changes. Outbound calls blocked. Banner on tenant dashboard.
- **Credit adjustment maker-checker:** Super Admin requests adjustment + reason → second admin approves → applied. Audit trail mandatory. Linked credit note generated.
- **Soft alerts cascade:** at 80%/90%/95% thresholds, alerts dispatched via email + WhatsApp + in-app. Visible in Alerts Center. Duplicate suppression within 24h.

CRUD Operations

Action	Trigger	Mock Behavior
--------	---------	---------------

Top Up via Stripe	Credit Purchase → Pay	Mock 3-second redirect → callback → credits added. Receipt emailed (mock toast).
Submit Offline Payment	Offline Payment → Submit	Status: Pending Approval. Routed to Finance queue.
Approve Offline Payment	Approval Queue → Approve (1st)	Routes to second approver.
Approve Offline Payment (2nd)	Approval Queue → Approve (2nd)	Credits added. Tenant notified.
Generate Invoice	Auto on cycle close OR Admin Manual	GST calculated. Number assigned. PDF generated (mock).
Trigger Hard Stop	Mock: balance → 0	Status flip. Tenant card change. Block all outbound. Alert.
Adjust Credits (Manual)	Admin → Credit Adjustment	Maker submits. Routes to Checker. On approval, applied.
Trigger Soft Alert	Mock: balance crosses threshold	Email + WhatsApp + in-app dispatched. Visible in Alerts Center.

Navigation Connections

- Sidebar (Tenant) → Billing section
- Sidebar (Super Admin) → Billing & Finance section
- Tenant detail → Billing tab
- Invoice → 'Download PDF' / 'Send by Email'
- Hard Stop banner → 'Top Up Now' CTA
- Approval queue badge in Super Admin sidebar

Demo Walkthrough

Tenant Admin tops up ₹50,000 via Razorpay. GST invoice generated with GSTIN, HSN, IGST/CGST/SGST. Watches credit balance increase. Two weeks later, hits 80% threshold → soft alert → ignores. Hits 95% → second alert → tops up via offline bank transfer → uploads proof → Finance Approver 1 → Approver 2 → credits added. Hits 100% in another scenario → Hard Stop active → outbound calls blocked → banner shown. Super Admin grants override (M15) for grace credits.

Existing Files (Reuse / Refactor)

src/app/pages/customer/BillingDashboard.tsx · CreditPurchase.tsx · UsageMetering.tsx · pages/admin/BillingDashboard.tsx · CreateInvoice.tsx · BillingRules.tsx · hooks/useBilling.ts (503 L). Offline payment + GST — major retrofit needed.

Module 15 — Admin & Super Admin Controls

Mission: Operational command center for platform operator. Kill-switches, overrides, impersonation, telephony provisioning. Every action 2FA-gated, mandatory reason, audit logged.

Pages to Build / Refactor

Page	Path	Status	Notes
Super Admin Dashboard	/admin/dashboard, /admin	Reuse	Platform health: uptime, error rates, active tenants, in-flight calls, revenue today, risk signals.
Super Admin Controls Hub	/admin/admin-controls, /controls	Refactor — full wiring	Cards for: Emergency Shutdown · Tenant Suspension · Override Permissions · Force Password Reset · Abuse Prevention · Data Access Override · Maintenance Mode.
Emergency Kill-Switch	(modal)	NEW (currently console.log)	Modal: scope (Platform-wide / Specific Tenant / Specific Campaign), mandatory reason, 2FA challenge, confirmation. On confirm: action executed (mock), audit logged, all super admins + affected tenants notified.
Tenant Suspension	(modal on tenant)	NEW (currently console.log)	Mandatory reason code + confirmation + 2FA. On confirm: status flip + active sessions terminated + campaigns paused. Audit logged.
Override Grant	/controls/overrides	NEW (currently Placeholder)	Wizard: pick tenant + override type (grace credits / trial extension / unlock / hard-stop bypass) + amount + duration (hard cap) + reason. Maker-checker required. Auto-revert on expiry.
Override List	/controls/overrides/active	NEW	Active overrides across all tenants. Per row: tenant, type, expiry. Revoke action.
Impersonation Sessions	/admin/impersonation	NEW	List active impersonation sessions. Per row: super admin, tenant, ticket reference, time remaining. Force-end action.
Impersonate User Modal	(modal on user detail)	NEW	Mandatory: support ticket reference + duration (max 2h). 2FA challenge. On confirm: session opens. Tenant audit log records the impersonation.

Impersonation Active Banner	(component)	NEW	When impersonating, prominent banner across screen: 'Impersonating [User] from [Tenant]. Session ends in 1h 23m.' End Session button always visible.
Abuse Prevention Controls	/controls/abuse-prevention	Reuse + wire	Rate limits per tenant, IP blocks, suspicious-activity rules. Currently UI-only — wire actions to mock store.
Telephony Provider Onboarding	/admin/telephony/providers/create	NEW (M13 cross-link)	Wizard: credentials + regions + rate card + test calls + promote.
Force Password Reset	(action on user)	NEW (currently console.log)	Mandatory reason. 2FA. Confirmation. Reset email sent (mock). Audit logged.
Maintenance Mode	/settings/maintenance	NEW	Toggle platform-wide or tenant-specific maintenance window. Banner shown. New logins blocked.
Audit Explorer (Super Admin View)	/controls/audit	NEW (currently Placeholder)	Cross-tenant audit search. Filters: actor type (user/AI/system), action, entity, tenant, date range. Export.
2FA Challenge Modal	(component, used everywhere)	NEW	Reusable: title, action description, 6-digit input, verify. Used for all destructive Super Admin actions.

Customizations & Special Behavior

- **ALL super admin destructive actions require:** (1) explicit confirmation modal · (2) mandatory reason in plain text · (3) 2FA challenge · (4) audit log entry · (5) notification to affected parties. Currently most are wired to console.log — this is the biggest fix in the module.
- **Kill-switch propagation:** platform-wide kill-switch terminates all in-flight AI calls within 500ms (mock toast 'Killed 247 active calls'). Tenant kill-switch terminates that tenant's calls only.
- **Time-bounded overrides:** every override has a hard maximum duration. Visible countdown. Auto-revert on expiry. Manual revoke option.
- **Impersonation hardening:** session duration capped at 2 hours. Tenant audit log shows impersonation prominently. Cannot modify billing, change passwords, or delete records during impersonation (UI gates).
- **Audit immutability:** even Super Admin cannot edit/delete audit log entries. Hard delete requires break-glass with separate approval (UI explains, doesn't allow).
- **Cross-tenant aggregate views:** Super Admin sees PII-safe aggregates only. To access individual tenant data, must impersonate (with audit).

CRUD Operations

Action	Trigger	Mock Behavior
Trigger Kill-Switch	Controls → Kill-Switch	Modal → reason → 2FA → confirm. Action executed. Audit + notifications dispatched.
Suspend Tenant	Tenant detail → Suspend	Reason → 2FA → confirm. Status flip. Sessions killed. Campaigns paused.
Grant Override	Controls → Overrides → Grant	Wizard. Maker submits. Routes to second admin. On approval, applied with countdown.
Revoke Override	Override list → Revoke	Reason. Reverts immediately.
Impersonate User	User detail → Impersonate	Ticket ref + duration + 2FA → session opens. Banner shown.
End Impersonation	Banner → End Session	Session ends. Audit logged in tenant log.
Force Password Reset	User detail → Force Reset	Reason + 2FA. Reset email sent (mock).
Onboard Telephony Provider	Telephony → Add Provider	Wizard → test calls → promote.
Activate Maintenance Mode	Settings → Maintenance → Activate	Window picker. Banner appears platform-wide. New logins blocked.

Navigation Connections

- Sidebar (Super Admin) → Controls section
- Tenant detail → 'Suspend' / 'Restore' / 'Lock' actions
- User detail → 'Impersonate' / 'Force Logout' / 'Force Reset' actions
- Audit Explorer is the universal back-reference for any privileged action

Demo Walkthrough

Super Admin notices in dashboard 'AI call success rate dropped 35% on Tenant X'. Opens controls. Decides to investigate. Impersonates Tenant Admin (ticket reference required, 2FA, 30min session). Sees configuration issue. Resolves. Ends session. Audit log shows impersonation event in BOTH super admin log and tenant log. Later: receives security alert. Triggers kill-switch on a single campaign (2FA, reason 'suspected abuse'). 247 calls terminated. Audit logged.

Existing Files (Reuse / Refactor)

src/app/pages/admin/SuperAdminControls.tsx · AbusePreventionControls.tsx. Most M15 — major fix needed (all wired to console.log).

Module 16 — Security, Compliance & Audit

Mission: Trust foundation. Cross-cutting layer every module honors: tenant isolation, encryption visibility, consent, DSR, regulatory mapping, AI guardrails, immutable audit log.

Pages to Build / Refactor

Page	Path	Status	Notes
Audit Log Viewer (Tenant)	/settings/audit, /team/audit	Reuse	Filter by actor (user/AI/system), event type, entity, date range. Export CSV.
Audit Log Viewer (Super Admin)	/controls/audit	NEW (M15 cross-link)	Cross-tenant search. Tamper-evidence indicator (hash-chain visualization).
Compliance Dashboard	/settings/compliance	NEW	Cards: Consent · Retention · DSR · Encryption Status · Backup Status · Recent Incidents. Each card actionable.
Consent Management	/settings/compliance/consent	Reuse + extend	Per-channel consent (call recording, marketing, cookies). Tenant-level + per-recipient.
Retention Policy	/settings/compliance/retention	NEW	Per-data-class retention windows (call recordings, messages, KB docs, audit logs). Within platform-defined maximums (cannot exceed).
DSR (Data Subject Request) Queue	/settings/compliance/dsr	NEW — CRITICAL	List of pending DSR. Per row: subject, type (Access/Rectification/Erasure/Portability/Restriction), request date, SLA deadline. Action: Process.
DSR Process Wizard	/settings/compliance/dsr/:id	NEW	Wizard: verify identity → search records → preview impact → execute (export / delete / restrict) → confirm completion. SLA timer visible (GDPR 30 days, DPDP per rules).
AI Guardrails Configuration	/settings/compliance/ai-guardrails	NEW	Tenant config: forbidden topics list, content-safety classifier sensitivity, PII redaction toggle. Test pane: input prompt → see if blocked.
AI Blocked Emissions Log	/settings/compliance/ai-blocked	NEW	When AI guardrail blocks an emission, logged here. Per row: agent, prompt, blocked content, reason, timestamp.

Incident Response	/settings/compliance/incidents	Reuse + wire	Active incidents. Severity. Affected tenants. Actions: Open Incident Mode, Generate Breach Notification.
Breach Notification Generator	/settings/compliance/incidents/:id/notify	NEW	Templated 72-hour notification per GDPR/DPDP. Affected parties recipient list. Mock dispatch.
Backup & DR Status	/settings/compliance/backup	NEW	Backup health (last snapshot, backup size, off-site replication). RTO/RPO indicators. Snapshots list. Restore wizard with dual approval.
Restore Wizard (Super Admin)	(modal)	NEW	Dual approval. Pick snapshot. Pick tenant. Mock 2-minute restore animation.
Encryption Status	/settings/compliance/encryption	NEW	Status display only: AES-256 at rest, TLS 1.3 in transit. Last key rotation. Mock — UI shows policy.
Tenant Isolation Verification	/admin/compliance/isolation	NEW	Super Admin tool: run mock cross-tenant query test. Shows '0 cross-tenant queries returned' (compliance check).
Password Policy	/settings/security/password-policy	Reuse + wire	Configurable: min length, complexity, rotation, reuse prevention, breach-checked on set.
Session Timeout Policy	/settings/security/sessions	NEW	Configurable: idle timeout, absolute timeout, force-logout schedule.

Customizations & Special Behavior

- **Audit log immutable + tamper-evidence:** viewer shows hash-chain check status per entry. 'Verified' badge if hash chain intact, 'Tamper Detected' if broken. Cannot edit/delete entries (UI). Hard delete = break-glass admin only.
- **DSR workflow:** 5 types per regulation (Access, Rectification, Erasure, Portability, Restriction). Each has dedicated wizard. SLA timer visible. Compliance trail logged.
- **AI guardrails:** forbidden-topic list editor. Content-safety sensitivity slider (Strict/Standard/Lenient). Tenant config + platform-mandatory base rules. Tenant CANNOT loosen base rules.
- **Retention enforcement:** tenant configures retention WITHIN platform maximums. Auto-purge schedules visible. Legal hold flags prevent purge for selected records.
- **Breach notification:** templated communication per GDPR/DPDP. 72-hour SLA timer. Affected parties identified. Dispatch via email + WhatsApp + in-app.
- **Backup + restore visibility:** RTO $\leq$ 4h / RPO $\leq$ 1h targets visible. Snapshot list with timestamps. Restore action (Super Admin only, dual approval).
- **Consent log per call:** every recorded call shows consent timestamp + jurisdiction flag. Click → consent disclosure transcript.

CRUD Operations

Action	Trigger	Mock Behavior
Submit DSR	Compliance → DSR → Submit	Subject info + type. Status: Open. Routed to processor.
Process Access DSR	DSR detail → Process → Access	Search records → preview → generate ZIP export. Mark complete.
Process Erasure DSR	DSR detail → Process → Erasure	Identity verification → impact preview → confirm → records erased (mock). Audit log entry.
Configure AI Guardrail	AI Guardrails → Add Forbidden Topic	Topic + reason. Tested in pane: input prompt → blocked / allowed.
Open Incident	Incidents → Open Incident	Severity + affected scope + initial findings. Routes to incident dashboard.
Generate Breach Notification	Incident → Generate Notification	Template wizard. Recipient list. Send mock dispatch.
Restore from Backup	Backup → Restore	Maker submits. Routes to checker. On approval, mock restore (2-min animation).
Configure Retention	Retention → Edit	Per data class. Within platform maximum. Save.

Navigation Connections

- Sidebar (Tenant) → Settings → Compliance
- Sidebar (Super Admin) → Compliance section
- Audit log entry → drill into record / actor / event
- DSR wizard → search records → preview → execute

Demo Walkthrough

Tenant Admin receives DSR via email (manual entry). Submits via DSR Queue with subject info + type 'Erasure'. Verifies identity (mock). Searches → 12 records found → preview → execute → records erased (lead, transcript, recording). Compliance Officer audits — sees DSR completed within SLA. Separately: Super Admin opens Incident Response → 'suspicious login pattern' → opens Incident Mode → reviews logs → generates breach notification (templated) → dispatches.

Existing Files (Reuse / Refactor)

src/app/pages/admin/PermissionAuditLog.tsx · settings/AuditCompliance.tsx · pages/customer/ComplianceConsent.tsx (827 L) · WhatsAppConsent.tsx. DSR + AI Guardrails + Backup + Incident — BUILD FROM SCRATCH.

6. Build Sequencing — Six Waves

The work is sequenced into 6 waves per PRD §8. Each wave produces a demoable increment. Do not start Wave N+1 before Wave N's modules have passed module-level acceptance (Section 3.1).

6.1 Wave 1 — Foundation (3 weeks)

Goal: make tenants creatable, users manageable, and customization possible. Without this, no other module is meaningful.

Module	Story Count	Key Deliverables
M1 Tenant & License	7 stories	Create Tenant w/ India KYC (already done) · License management w/ maker-checker · Tenant detail w/ tabs · Hard-stop indicators · Audit visibility
M2 User, Role & Permissions	7 stories	Metadata-driven permission tree · 2FA enrollment wizard · Session management · Force-logout · AI vs Human distinction
M3 Customization & Metadata Platform	7 stories	Custom object builder · Field builder (20+ types) · Formula engine · Validation rules · Layout editor per role · List view builder · Custom App builder · Schema viewer
Mock store foundation	Cross-cutting	Centralized Zustand store · API simulation layer · 16 slices · Seed data (12 tenants, 200 leads, etc.)
i18n setup	Cross-cutting	react-i18next wired · English strings externalized · placeholder for Hindi

Wave 1 Risks

- M3 is the largest single piece of work in the entire program. It will likely take 2 of the 3 weeks. Pair on it.
- Once M3 ships, M2's permission tree must be refactored to consume M3 metadata — DO NOT skip this.
- Mock store foundation MUST land first. Do not let any team ship a feature with local useState data.

6.2 Wave 2 — CRM Core (2 weeks)

Goal: the CRM works end-to-end. Lead → Deal → Won.

Module	Story Count	Key Deliverables
M4 Lead Management	7 stories	M3-driven lead UI · CSV import wizard · Lead source webhook configuration · Refactor timeline w/ AI badges · Reuse existing scoring + dedup

M5 Deal & Pipeline	7 stories	Build forecasting (currently Placeholder) · Build attribution tab · AI advisory side panel · Mandatory close reason · Discount approval workflow
M13 (partial: telephony + webhooks)	Cross-link	Number inventory · Provider onboarding · Outbound webhook config w/ HMAC

6.3 Wave 3 — Engagement Channels (3 weeks)

Goal: all four engagement channels operate at parity. India compliance for SMS.

Module	Story Count	Key Deliverables
M6 AI Calling	7 stories	Polish existing 12 pages · Add call state machine visualization · Live consent capture flow · Cost safeguard explicit cards
M7 WhatsApp	7 stories	Polish 12 pages · AI ↔ Human ownership transitions · Take-over / Hand-back flows · Platform-mediated Meta API framing
M8 SMS	5 stories	BUILD DLT Registry (CRITICAL India compliance) · Quiet hours / DND enforcement · Transactional vs Promotional category · Keyword routing config
M9 Email	5 stories	BUILD Deliverability dashboard · BUILD Suppression list · Polish AI draft mode · Public unsubscribe page
M13 (completion: calendar + lead sources)	Cross-link	Google Calendar + Outlook OAuth · Meta Lead Ads · LinkedIn · IndiaMART · Google Ads · Shopify · WooCommerce inbound webhooks

6.4 Wave 4 — Orchestration & Commercials (2 weeks)

Goal: campaigns can orchestrate multi-channel sequences. Billing reflects India GST.

Module	Story Count	Key Deliverables
M10 Campaigns & Workflow	6 stories	Polish existing node builder (good foundation) · BUILD A/B testing · Workflow versioning · Pause vs Stop semantics · Audience builder using M3
M14 Billing & Usage	6 stories	SWAP NPR → INR · BUILD GSTIN + HSN/SAC + IGST/CGST/SGST invoicing · BUILD offline payment + dual approval · Hard-stop propagation · Monotonic numbering · Credit adjustment maker-checker

6.5 Wave 5 — Intelligence & Growth (2 weeks)

Goal: growth surfaces functional. Custom analytics live.

Module	Story Count	Key Deliverables
M11 Business Playground	6 stories	Reuse existing 9 pages · Unify credit-preview confirmation pattern · Visible scraping allowlist · OAuth connection manager
M12 Analytics & Reporting	5 stories	Reuse existing report builder · BUILD custom dashboard builder (drag-drop tiles) · BUILD anomaly insight cards · Report scheduling · M3-driven sources

6.6 Wave 6 — Platform Operations (2 weeks)

Goal: Super Admin can run the platform safely. Compliance posture defensible.

Module	Story Count	Key Deliverables
M15 Admin Controls	5 stories	WIRE all kill-switch handlers (currently console.log) · Confirmation + 2FA + reason on every action · BUILD impersonation w/ time-bound + audit · BUILD telephony provider onboarding
M16 Security & Compliance	5 stories	BUILD DSR workflow (5 types) · BUILD AI guardrails configuration · BUILD incident response · BUILD breach notification · BUILD backup + restore w/ dual approval · Audit tamper-evidence indicators

6.7 Effort Distribution

Relative effort across waves to inform staffing:

Wave	Module Coverage	Relative Effort	Critical Path
Wave 1 — Foundation	M1, M2, M3, store, i18n	30%	M3 is the program's biggest single deliverable
Wave 2 — CRM Core	M4, M5, partial M13	12%	Mostly polish on existing — fast wave
Wave 3 — Engagement	M6, M7, M8, M9, M13	20%	DLT (M8) and Calendar (M13) are net-new
Wave 4 — Orchestration	M10, M14	15%	GST swap is invasive across many invoice screens
Wave 5 — Intelligence	M11, M12	10%	Dashboard builder is the new piece
Wave 6 — Platform Ops	M15, M16	13%	DSR + AI guardrails + backup are all new

7. Cross-Cutting Engineering Checklist

These items apply across every module. Treat as definition-of-done at the program level. Audit each PR against this list.

7.1 Mock Store Hygiene

- Every entity read/write goes through the centralized store. No useState for cross-page data.
- Cross-references use IDs. Resolution happens at the store level.
- Every privileged write emits an audit log entry automatically (no manual call needed in components).
- Latency simulation honored on every API call (300–800 ms reads, 600–1500 ms writes).
- 5% failure injection respected — UI shows retry-friendly errors.

7.2 Permission Enforcement

- Every action button consults usePermissions(action, scope). Disabled (with tooltip) if denied.
- Sidebar items hidden if user lacks any permission to that section.
- Page-level guard: redirect to /403 if user lacks read access to the page.
- Auditor role: see-only across the platform. No edit / delete / create buttons render.

7.3 India Compliance Surfaces

- INR, IST, en-IN as default tenant locale. NPR / Asia/Kathmandu / ne-NP retained as options.
- PAN format validated everywhere asked: `^[A-Z]{5}[0-9]{4}[A-Z]$`
- GSTIN format validated: `^[0-9]{2}[A-Z]{5}[0-9]{4}[A-Z]{1}[1-9A-Z]{1}Z[0-9A-Z]{1}$`
- Indian mobile validated: `^[6-9]\d{9}$`
- Indian PIN validated: `^[1-9][0-9]{5}$`
- Invoices show GSTIN, HSN/SAC, IGST/CGST/SGST breakdown for India tenants.
- SMS templates show DLT registration status. Send disabled if unregistered.
- DSR workflow available in tenant compliance settings.

7.4 Form Validation Standards

- Every required field shows red asterisk.
- Inline error message under field (not toast) on blur.
- Submit blocked while errors present. Scroll-to-first-error on attempt.
- Cross-field validation (e.g., end date > start date) shown at the section level.
- Use react-hook-form + zod for new forms. Migrate existing forms when refactored.

7.5 Loading, Error, Empty States

- Skeleton loader on initial page load.
- Spinner on action buttons during submission. Button disabled.
- Error state: friendly message + retry button. Never raw error JSON.
- Empty state: illustration + headline + 'Create First X' CTA.
- Error toasts auto-dismiss in 5s. Success toasts in 3s. Critical errors persist until dismissed.

7.6 Audit Logging

- Every CRUD on entities tagged 'auditable' writes a log entry: { actor, actorType, action, entity, entityId, before, after, reason, timestamp }.
- Privileged actions (suspend, override, kill-switch, impersonate, restore, force-logout) require reason at UI layer.
- Audit log entries display actor type icon: human / AI / system.
- Audit Log Viewer is the universal back-reference. Every entity's detail page has 'View Audit' link.

7.7 AI Attribution

- Every AI-generated event displays an AI badge with Agent ID + Campaign ID.
- AI actions never linked to human users — actor is the AI agent.
- Lead/Deal timelines visually distinguish AI events (different background color).
- AI advisory suggestions on deals show explicit 'Execute' / 'Dismiss' buttons. Never auto-apply.

7.8 Maker-Checker Pattern

- Reusable ApprovalRequest component with states: Pending → Approved → Executed (or Rejected).
- Maker submits, sees pending status. Cannot approve own request.
- Checker sees in approval queue (filtered to their permission scope). Approves or rejects with comment.
- Both approvers logged. Approval queue badge in sidebar shows pending count.
- Applies to: license assignment, credit adjustment, offline payment, override grant, suppression removal, backup restore.

7.9 Routing & Navigation

- All routes use kebab-case. Standardize current /deals/list and /deals (one canonical).
- Every page has breadcrumb in TopBar reflecting hierarchy.
- Browser Back button always works (no React state-only navigation).

- Deep links to entity detail pages are shareable: `/:entity/:id` renders correctly from cold start.
- Zero Placeholder routes by Wave 6 end.

7.10 Performance Smell Tests

- List pages render < 1s with 200 records (mock pagination on 25).
- Form pages render < 500ms.
- Builder canvases handle 50 nodes without lag.
- Live monitor pages update at 1 Hz max.
- Lighthouse Performance score ≥ 75 on production-like build (post-backend, will tune further).

7.11 Responsive Design

- Desktop-first design (1280px primary).
- Tablet (1024px): 2-column $\rightarrow$ 1-column where needed; sidebar collapsible.
- Mobile (768px): single-column; sidebar drawer; tables become card lists.
- Builder canvases (workflow / flow / layout) are desktop-only — show 'use desktop' message on mobile.

8. Per-Page Status Tracker (High Level)

Master sheet — every page in the v0 codebase mapped to its target status. Use the accompanying XLSX (OmniCRM_UserStories_Linear.xlsx) for ticket-level tracking.

8.1 Reuse-As-Is (~45 pages)

These pages have strong existing UI and require only mock-store wiring + minor polish.

Module	Pages
M1 Tenant	TenantList · TenantDetail · TenantEdit · CompanySettings
M2 Users	RolesList · UserRoleAssignment · TenantRolesList
M4 Leads	LeadKanban · LeadScoringDashboard · LeadAssignmentConfig · DuplicateLeadManagement
M5 Deals	DealDashboard · DealList · DealPipeline · DealProgressionReport
M6 AI Calling	AICallingDashboard · LiveMonitor · CallLogs · CallDetail · TransferQueue · CallAnalytics · CallLibrary · CallSchedulingRetry · CostTransparencyReport · InboundCallSetup
M7 WhatsApp	WhatsAppDashboard · WhatsAppSetup · WhatsAppLeads · WhatsAppAnalytics · WhatsAppSettings
M8 SMS	SMSDashboard · SMSInbox · SMSAnalytics · SMSSettings · SMSShortLinks
M9 Email	EmailDashboard · EmailCompose · EmailSignatures · EmailAnalytics
M11 Playground	BusinessPlayground · ProductResearch · KnowledgeBase · LeadGeneration · SocialMediaMarketing · StrategyPlanning · ResponseManagement · AssetLibrary · Storage
M12 Analytics	ReportsAnalyticsDashboard · AICallingReport · CRMReport · CampaignsReport · DealsReport · PerformanceReport · CostReport

8.2 Major Refactor (~85 pages)

Existing UI shell is reusable but requires significant logic changes for India compliance, M3-driven UI, validation, or workflow.

Module	Pages
M1 Tenant	CreateTenant (DONE) · LicenseManagement · LicenseDetail · SubscriptionPlans (INR) · UsageAnalytics
M2 Users	CreateEditRole (M3-driven) · PermissionMatrix (M3-driven) · TenantCreateEditRole · TenantPermissionMatrix
M4 Leads	LeadManagement (M3-driven) · LeadDetail (M3-driven) · CreateLead (M3-driven)

M5 Deals	CreateDeal (approval + close reason) · DealDetail (insights, attribution)
M6 AI Calling	EditAgent · ConversationFlowEditor · ComplianceConsent · CostGovernance · PromptTemplateEditor
M7 WhatsApp	WhatsAppInbox · WhatsAppTemplates · CreateWhatsAppTemplate · WhatsAppAutomation · WhatsAppConsent · CreateWhatsAppCampaign · CreateWhatsAppCampaignAdvanced · WhatsAppAPI
M8 SMS	CreateSMSCampaign · SMSTemplates
M9 Email	EmailInbox · CreateEmailCampaign · EmailTemplates · CreateEmailTemplate · EditEmailTemplate · EditEmailCampaign · EmailSettings
M10 Campaigns	CampaignManager · CreateCampaign · CreateOmniCampaign · WorkflowAutomation · CampaignBudgetManager · CampaignExecutionMonitor
M12 Analytics	CustomReportBuilder (M3-driven, scheduling)
M14 Billing	BillingDashboard (tenant + admin) · UsageMetering · CreditPurchase · InvoiceList · InvoiceDetail · CreateInvoice · CreditConfiguration · CreditManagement · BillingRules · AlertsCenter · Subscription
M15 Controls	SuperAdminControls · AbusePreventionControls
M16 Compliance	PermissionAuditLog · TenantPermissionAuditLog · AuditCompliance · SecurityAuth

8.3 Build New (≈40 pages)

Module	Pages
M1 Tenant	License Approval Queue · Temp License + Payment Proof
M2 Users	User Detail · Active Sessions · 2FA Enrollment Wizard
M3 Customization (entire module)	Customization Hub · Object List · Object Builder · Object Detail · Field Builder · Formula Field Builder · Validation Rule Builder · Layout Editor · Custom App Builder · Custom Apps List · Schema Viewer · Metadata Version History
M4 Leads	CSV Import Wizard · Lead Sources Hub · Lead Source Configuration
M5 Deals	Deal Edit (currently Placeholder) · Deal Forecasting (currently Placeholder) · Companies · Contacts · Discount Approval Queue
M8 SMS	DLT Registry · Submit Template to DLT · Keyword Routing Configuration
M9 Email	Deliverability Dashboard · Suppression List · Public Unsubscribe Page
M10 Campaigns	Campaign Detail (currently Placeholder) · Campaign Edit · Campaign Analytics · Workflow List · A/B Test Configuration · A/B Test Results · Audience Builder modal
M11 Playground	OAuth Connection Manager

M12 Analytics	Custom Dashboard Builder · My Dashboards · Dashboard View · Anomaly Insights Center · Scheduled Reports
M13 Integrations	Integrations Hub · Telephony Number Inventory · Telephony Provider Management · Calendar Connection · Outbound Webhooks (refactor to detail) · Inbound Lead Sources · Lead Source per type · Accounting Export
M14 Billing	Payment Methods · Offline Payment Proof Upload · Offline Payment Approval Queue · Credit Adjustment (Super Admin)
M15 Controls	Override Grant · Override List · Impersonation Sessions · Audit Explorer · Maintenance Mode
M16 Compliance	Compliance Dashboard · Retention Policy · DSR Queue · DSR Process Wizard · AI Guardrails Configuration · AI Blocked Emissions Log · Breach Notification Generator · Backup & DR Status · Encryption Status · Tenant Isolation Verification · Session Timeout Policy

9. Sprint Plan Template (2 Engineers + 1 Designer)

Suggested allocation. Adjust to your team. Two-week sprints. Designer 0.5 FTE on this program (designs flow ~1 sprint ahead of engineers).

9.1 Sprints 1–2 (Wave 1A — Foundation)

Engineer 1	Engineer 2	Designer
Mock store: scaffold, all 16 slices, seed data, API simulation, audit auto-write	i18n setup. Refactor M1 (license, suspend, restore, approval queue, temp license)	M3 designs: Object Builder, Field Builder, Formula Editor
M2 refactor: metadata-driven permission tree, 2FA enrollment, sessions, force-logout	M1 polish + first M3 page (Customization Hub, Object List)	M3 designs: Layout Editor, Custom App Builder, Schema Viewer

9.2 Sprints 3–4 (Wave 1B — M3 Big Push)

Engineer 1	Engineer 2	Designer
M3: Object Builder · Field Builder (10 simple types) · List View Builder	M3: Formula Builder · Validation Builder · Layout Editor	M4 designs: CSV Import wizard, Source connectors. M5 designs: Forecasting, Attribution
M3: Field Builder (10 advanced types incl. Lookup, Master-Detail, Auto-Number, Geolocation) · Schema Viewer	M3: Custom App Builder · Apps List · Version History · Recycle Bin	M6/M7 polish designs. M8 DLT designs.

9.3 Sprints 5–6 (Wave 2 — CRM Core)

Engineer 1	Engineer 2	Designer
M4 refactor: M3-driven Lead UI · CSV Import wizard · Lead Sources Hub	M5 build: Forecasting page · AI Insights panel · Attribution tab · Discount approval	M9/M10/M14 designs (deliverability dashboard, A/B tests, GST invoice)
M5 polish: Close reason, lock-on-close, Companies/Contacts pages	M13 partial: Number inventory, provider management, outbound webhook detail	M11/M12 designs (dashboard builder, anomaly insights). M15 designs (impersonation, overrides)

9.4 Sprints 7–8 (Wave 3 — Engagement)

Engineer 1	Engineer 2	Designer
------------	------------	----------

M6 polish: state machine viz, live consent, cost cards · M7 polish: take-over flow	M8 build: DLT Registry · Submit Template · Quiet hours · Keyword routing	M16 designs (DSR wizard, AI guardrails, breach notification)
M9 build: Deliverability dashboard · Suppression list · Public unsubscribe · AI draft mode	M13 build: Calendar OAuth · Meta Lead Ads · LinkedIn · IndiaMART · Google Ads · Shopify · WooCommerce · Generic webhook	Refinement designs based on testing feedback

9.5 Sprints 9–10 (Wave 4 — Orchestration & Commercials)

Engineer 1	Engineer 2	Designer
M10 polish: existing node builder · Build A/B testing · Workflow versioning · Pause/Stop semantics	M14 swap NPR → INR everywhere · GST invoicing · Offline payment + dual approval · Hard-stop propagation	Polish + accessibility audit designs

9.6 Sprints 11–12 (Wave 5–6 — Intelligence + Platform Ops)

Engineer 1	Engineer 2	Designer
M11 polish: credit preview unified · M12 build: Dashboard builder · Anomaly insights · Scheduled reports	M15 wire: kill-switch · 2FA gating · Reason capture · Impersonation · Telephony provider onboarding	Demo polish, microcopy review
M16 build: DSR workflow · AI guardrails · Incident response · Breach notification · Backup/restore	M16 build: Audit tamper-evidence · Retention policy · Encryption status · Cross-tenant aggregate views	Final QA pass

Note: This is a calendar-driven plan. The actual sequencing should follow dependency, not date. Engineering Manager owns reprioritization week-to-week.

10. Open Questions for Stakeholders

These decisions affect frontend behavior. Resolve before the relevant wave begins.

Q1 — Multi-tenancy isolation symbol

Question: When a Super Admin browses tenant data (without impersonation), should aggregate views show tenant names or anonymized IDs (TNT-001 etc.)?

Default if unanswered: Show tenant names — Super Admin needs operational context. Toggle to anonymize for screenshots / demos.

Q2 — Custom object record IDs

Question: Format for custom object record IDs? Salesforce uses 15/18-char alphanumeric. Zoho uses long numerics. Should each object have a configurable ID prefix (DST-XXXXX for Distributor, ORD-XXXXX for Order)?

Default: Object-prefix + 6-digit zero-padded counter (DST-000001, DST-000002...). Configurable in Object Builder.

Q3 — Workflow node node-type extensibility

Question: Will tenants be able to define custom Action nodes (e.g., 'Call External API') in M10? Or are nodes a closed set defined by platform?

Default: Closed set in v1. Tenants extend via 'Webhook Action' node. Custom node SDK is a v2 consideration.

Q4 — Customer Portal (Tier 3) scope

Question: How extensive is Customer Portal in v1? Just account view + ticket creation? Or full distributor workflows (place order, view inventory)?

Default: Minimal v1: dashboard + tickets + documents + profile. Distributor-specific workflows are tenant-customized via M3 Custom Apps.

Q5 — Color/branding customization per tenant

Question: Can tenants customize their UI brand colors / logo / favicon? White-labeling extent?

Default: Tenant logo + 1 primary color in v1 (replaces brand-blue accent). Full white-label is Enterprise tier only.

Q6 — Currency on cross-currency tenants

Question: If a tenant operates in multiple currencies (e.g., India INR + UAE AED), how does invoicing work? Per-deal currency? Fixed per tenant?

Default: Tenant has primary currency. Per-deal currency override permitted. Invoice generated in deal currency. Reporting consolidates to tenant primary using mock FX rate.

Q7 — AI agent voice catalog

Question: Voice options for AI agents? Open-ended TTS provider list, or fixed catalog?

Default: Fixed catalog of 20 voices (Hindi M/F, English Indian M/F, etc.) per platform. Voice selection in agent config.

Q8 — Maker-checker — same-team rule

Question: Can the maker and checker be from the same tenant team, or must checker be from a separate Finance / Compliance team?

Default: Same-team allowed but two distinct users required. Compliance-grade actions (DSR erasure, backup restore) require checker from a designated Compliance role.

Q9 — Offline mode

Question: Any offline-first requirements? Service workers? Or always-online?

Default: Always-online for v1. Connection-loss banner with retry. Offline-first deferred to mobile app phase.

Q10 — Demo seed data persistence

Question: When demoing on a fresh browser session, should mock data persist (localStorage) or reset on every reload? Both useful for different scenarios.

Default: Reset on reload by default. Localstorage toggle in dev menu for persistent demo sessions.

11. Closing Notes

11.1 Companion Artifacts

- **OmniCRM_Platform_PRD_v1.0.docx** — the source of truth. Read it. Refer to its module sections, user stories, acceptance criteria. This brief is the implementation guide; the PRD is the contract.
- **Nightangle_CRM_Gap_Analysis_v1.0.docx** — per-module audit of what exists, what's partial, what's missing. Use it to scope each story.
- **OmniCRM_UserStories_Linear.xlsx** — ticketing. Import to Linear / Jira / Asana. One row per user story. AC sheet has 485 rows for sub-issues.
- **Sample fixed file CreateTenant.tsx** — US-M1.1 reference implementation, ~1,231 lines. Demonstrates: validation patterns, document versioning, async provisioning state machine, India compliance, success state design. Use as template for similar wizards.

11.2 Engineering Principles to Hold

- **The mock store is sacred.** Every PR that adds local state for shared data is rejected. The whole point is backend-swap parity.
- **M3 is the spine.** If you find yourself hardcoding fields or layouts anywhere, stop and consume from M3 instead.
- **Acceptance criteria are non-negotiable.** Every AC = a clickable demonstration in the running build. PR review verifies.
- **India compliance is not a bolt-on.** GSTIN, DLT, DPDP, PAN format are first-class concerns from day one.
- **Audit, audit, audit.** Every privileged write logs. The Audit Log Viewer is the program's QA tool.
- **Don't build for backend you don't have.** If a flow needs server logic the mock can't simulate, surface the limitation in PR review and propose a UI compromise.

11.3 Final Acceptance

This brief is closed when the demo script in Section 3.3 runs cleanly on the mock-data build, all 97 user stories are marked done in the tracker, and the PRD's acceptance criteria can each be demonstrated in under 30 seconds. At that point, the frontend is ready for backend integration. The mock store becomes 'demo mode' and real APIs replace api.ts.

Good building.